

Natural Tooth Bleaching

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History of Bleaching

For well over a century, bleaching in some form has been used to achieve a lighter and more desirable tooth color. Dental journals in the latter half of the 19th century frequently contained articles on the efficacy of bleaching teeth with different techniques.[1–3] The safety of the procedure was well investigated, and most of the bleaching agents were considered to be safe by the standards of the time. Although the process was thought to be time-consuming and relapse was considered a consistent problem, bleaching was well accepted by many dentists.[4–6] The following is a brief history of dental bleaching approaches.

By the mid-1800s, the in-office bleaching agent of choice for nonvital teeth was chloride of lime.[7–9] Around that time, Truman introduced chlorine from a calcium hydroxide and acetic acid solution for bleaching nonvital teeth; this was supplied commercially as a liquid chloride of soda.[5] Other agents used in the 1800s for nonvital tooth bleaching included aluminum chloride,[10] oxalic acid,[3,11] Pyrozone

(ether peroxide),[2] hydrogen dioxide (hydrogen peroxide or perhydrol),[12] sodium peroxide,[12] sodium hypophosphate,[8] and cyanide of potassium. The active ingredient common to all of these was an oxidizing agent that acted either directly or indirectly on the organic portion of the tooth. Sulfurous acid, in contrast, achieved results as a reducing agent.[13,14] Generally, the most effective yet safe bleaching agents were direct oxidizers or an indirect oxidizer such as a chlorine derivative.[6,15] However, the choice of bleaching agent depended primarily on the stain being removed. Iron stains were removed with oxalic acid, silver and copper stains with chlorine, and iodine stains with ammonia.[14–17] Metallic stains (such as those from amalgam) were considered the most resistant to bleaching. Concern about the effect of some of these bleaching agents on the teeth, tissues, and health of the patient was raised because certain agents, such as cyanide of potassium, were very poisonous[12] while others were highly caustic.

In the 1890s, the early emphasis was on bleaching nonvital teeth. However, by this time, a 3% solution of Pyrozone was being used safely as a mouthwash by both children and adults to reduce caries and to whiten the teeth.[18] A 5% solution proved to be safe and effective, but a 25% solution was caustic, causing tissue burns.[18] By 1910, an "in-office bleaching technique" similar to that used today, where high concentrations of hydrogen peroxide are used on well-isolated teeth, occasionally assisted by heat or light, was well established.[19]

Over the next few decades, the bleaching agents varied. By the 1940s, hydrogen peroxide and ether were used for vital teeth,[20] while, by the late 1950s, Pyrozone and sodium perborate were used internally for nonvital teeth.[21,22] In the late 1960s, the current "home bleaching technique," employing a custom-fit tray containing 10% carbamide peroxide solution, was first used by Klusmier.[23] However, the profession did not embrace the concept until it was described in an article published in 1989.[24]

In 1989, Haywood and Heymann wrote the first article on an at-home bleaching technique employing a custom-fit tray containing 10% carbamide peroxide solution,[24] and its publication coincided with the market introduction of carbamide peroxide as a bleaching agent.[25,26] This ushered in the current technique for at-home bleaching. Various chemical combinations have been tried. The effective ones all release oxidizing agents, whether supplied as hydrogen peroxide, carbamide peroxide, sodium percarbonate, or some other compound.[27] Following the popularity of the at-home bleaching by trays, there was a resurgence of in-office bleaching, with various light and chemical enhancements, as well as a proliferation of over-the-counter products. The terms *bleaching* and *whitening* are often used

interchangeably, and *lightening* is also used occasionally. There is not a clear distinction among the terms, although cosmetic products typically use the word *whitening* for the removal of surface stains, and true bleaching products, which change the inherent color of the tooth, are referred to as *bleaching* products.

Types and Nature of Stains/ Discolorations

Many types of color problems affect the appearance of the teeth. The cause of these problems varies, so the speed or efficacy with which they may be removed also varies. Discolorations may be extrinsic or intrinsic. *Extrinsic stains* are located on the surface of the tooth and are most easily removed by external cleaning. *Intrinsic stains* are located within the tooth and are accessible only by bleaching. Some extrinsic stains that remain on the tooth for a long time become intrinsic. Extrinsic color changes may be the result of poor oral hygiene. Intrinsic color changes may be caused by aging, microcracks in the enamel, tetracycline medication, excessive fluoride ingestion, severe jaundice in infancy, porphyria, dental caries, restorations, and thinning of the enamel layer, as well as ingestion of chromogenic foods and drinks and tobacco use. Other, less common medical situations and conditions may also cause the loss of desirable tooth color.

The causes of staining need to be assessed carefully to better predict the rate and degree to which bleaching will improve tooth color, because some stains are more responsive to the process than others.[24,28] For instance, the yellow discoloration of aging responds quickly to bleaching in most cases,[29] whereas a blue-gray tetracycline stain is tenacious and is seldom totally removed, only lightened.[26] In general, tetracycline-stained teeth are the slowest to respond to bleaching[30] (Fig 16-1). Brown (Fig 16-2a) or brown-fluoresced (Fig 16-2b) teeth are moderately responsive, and teeth discolored by age, genetics, smoking, or coffee are the fastest to respond.[31,32] White spots are not removed by bleaching but may be less noticeable when the remainder of the tooth is lighter.[33] By recognizing the likely cause of the stain, the dentist can better tell the patient the rate at which the teeth may lighten in color and the limits on the amount of improvement that can be expected.



Fig 16-1 Teeth demonstrating moderate, grayish tetracycline stain are excellent candidates for external bleaching but may take longer to respond.



Fig 16-2*(a)* Stain of unknown origin creates disfigurement in a 10-year-old boy with brownish areas surrounded by whiter areas. Brown stains are very amenable to bleaching treatments. *(b)* High fluoride content in drinking water can cause brown and white stains. The brown stain can be removed, but the white remains unchanged.

Discoloration from drug ingestion may occur either before or after the tooth is fully formed. Tetracycline is incorporated into the dentin sometime during tooth calcification (see Fig 16-1), probably through chelation with calcium, forming tetracycline orthophosphate.[34–36] There are several variations of tetracycline, and each derivative produces a different color in the tooth. Some teeth may be "banded" from the ingestion of different derivatives of tetracycline. When the teeth are exposed to sunlight, they become darker, with a distinct gray/blue-gray tinge. The teeth not exposed to the sunlight (ie, molars) do not darken to the same degree but remain more yellow in color. Tetracycline also has been reported to discolor fully formed, erupted permanent teeth. This discoloration is most often associated with minocycline, a drug commonly used in the treatment of acne.[37–39] The primary route of deposition is thought to be in the secondary dentin, although some reports suggest a staining similar to that of iron deposition.[40] Other antibiotics may also interact with calcium, iron, or other elements to form insoluble complexes that stain teeth.[41]

Excessive fluoride in drinking water, greater than 1 to 2 ppm, can cause metabolic alteration in the ameloblasts, resulting in a defective matrix and improper calcification of teeth.[42] An affected tooth shows a hypomineralized, porous subsurface enamel and a well-mineralized surface layer. These teeth have a glazed surface and may be very white except for areas of yellow, brown (see Fig 16-2b), or even black shading.

Some systemic conditions can cause tooth discoloration. Severe jaundice leads to staining by bilirubin. Erythroblastosis fetalis may also stain the teeth because of the destruction of red blood cells. Porphyria, a rare condition, manifests with purplish-brown teeth.

Aging is a common cause of discoloration. Over time, the underlying dentin tends to darken from the formation of secondary dentin, which is darker and more opaque than the original dentin. This occurs while the overlying enamel becomes thinner, a combination that often produces distinctly darker teeth. Additionally, chromogenic foods gradually build up stains in the enamel.

Dental caries produces varying stains depending on its stage of progression. Examples of caries-induced discolorations include an opaque white "halo," a grayish tinge, or a brown to black stain. These stains arise from the bacterial degradation of food debris. Metallic restorations, most notably dental amalgam, may cause a distinct staining of the tooth in addition to the shadow they may cast through adjacent enamel walls.

Another cause of discoloration to the tooth is trauma. The tooth may remain vital but can discolor, either from iron-containing hemoglobin in blood seeping into the dentinal tubules or from calcific metamorphosis (Fig 16-3). *Calcific metamorphosis* is the process by which the tooth deposits darker yellow secondary dentin in the pulp chamber, which may partially or completely obliterate the pulp chamber. Other more serious issues that may cause discoloration are a necrotic pulp that has resulted in a periapical radiolucency or a tooth that is undergoing internal or external resorption. Often in these cases, there is no pain, no swelling, no mobility, or any other symptom other than the onset of tooth darkening. Occasionally, caries and the resultant deep restoration may eventually result in pulpal death. A radiograph should always be taken of a single dark tooth, and pulp testing may also be indicated.

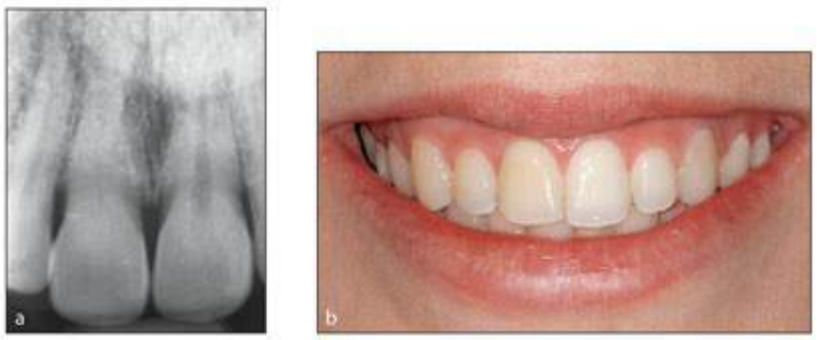


Fig 16-3*(a and b)* Calcific metamorphosis, seen in the maxillary right central incisor, is the condition in which the pulp chamber has been obliterated by secondary dentin. The tooth may remain vital but is not a good candidate for endodontic therapy and internal bleaching. External bleaching of the single dark tooth is best performed with tray application of the bleaching agent.

Indications for Bleaching

The primary indication for bleaching is patient dissatisfaction with tooth color,

either of all the teeth or teeth adjacent to restorations. While the source of the discoloration affects the degree of success and the rapidity with which it can be eliminated or minimized, it has been shown that even the most persistent discolorations can be lightened if the treatment is sufficiently extended.[43] Bleaching may be done in lieu of bonded resin composite restorations, porcelain veneers, or crowns to improve the tooth color. Patients may be satisfied with the results of bleaching such that more invasive treatment is not needed.[44] Even if porcelain or composite veneers are to be placed, the lighter color of the bleached teeth allows lighter and more translucent veneers, enhancing the natural appearance. Other indications include extending the esthetic life of existing crowns that are lighter than the adjacent natural teeth by returning the color of the natural teeth to the shade of the crown, or whitening a single dark tooth to the desired shade to match the adjacent teeth.

There is also a psychologic benefit of bleaching, especially in the 10- to 14-year age group, when discolored teeth are more likely to have an adverse impact on an individual's self-image. Bleaching removes a source of embarrassment, which may affect personality development during a child's middle school years.[45] Hence, the age of a bleaching patient can range from 10 years to 80 years or older. Patients tend to look most natural when the color of the teeth matches the white of the sclera in their eyes.[46] Therefore, a desired endpoint for bleaching is not the number of shade guide changes but a natural, beautiful appearance where the white of the teeth matches the white of the eyes.

Contraindications for Bleaching

Although bleaching is a safe and effective aid in improving the appearance of the teeth after a proper examination, not every discolored tooth requires bleaching. Superficial or extrinsic stains may be completely removed by a rubber cup with prophylaxis paste or by light abrasion with a rotary polishing device. The removal of discolored carious tooth structure and/ or a dark restoration and placement of a tooth-colored material may well make a marked improvement in the appearance of a tooth.

There are few contraindications for bleaching (Box 16-1), whether in-office or at-home dentist-prescribed techniques are being considered. In-office bleaching is not recommended for children with large pulps or teeth with cracks because of the higher concentrations of bleaching solutions used and subsequent sensitivity issues.

Likewise, exposed root surfaces and severe loss of enamel may be contraindications. At-home bleaching is generally not indicated for pregnant women or patients allergic to the ingredients in the carbamide peroxide preparations. Although there is no evidence that bleaching is harmful to the fetus or to infants, it has been recommended that pregnant and lactating women avoid bleaching because of the increased potential for gingival irritation.[47,48] Also, the mother's peace of mind should be taken into account so that she does not improperly relate bleaching problems to any issues that might occur with the birth or the baby. Because there is some evidence that peroxides may enhance the effect of known carcinogens, it may be prudent to have the patient forego tobacco use during the period of the bleaching process.[31,49,50] In one study of at-home bleaching, Nathanson and Parra[51] determined that there was no noticeable difference in the sensitivity reported by young patients compared with sensitivity reported by older patients, so larger pulp size may not be a factor for at-home bleaching. In a study of at-home bleaching with carbamide peroxide, Leonard et al[33] determined that there were no predictors of which individuals would experience sensitivity other than a history of sensitive teeth and more than one bleach application per day.[52] All other characteristics, such as pulp size, exposed dentin, cracks, gingival recession, caries, sex or age of the patient, or other physical characteristics, were not predictive of tooth sensitivity.

Box 16-1	**Contraindications for bleaching**

- Unrealistic expectations
- Unwillingness to comply with treatment
- Extensive existing tooth-colored restorations that do not need replacement or patient is unwilling to replace
- Inability to tolerate taste (at-home bleaching) or technique (in-office)
- Pregnancy or lactation

There are some potential bleaching situations with guarded prognoses (Box 16-2). Patients with hypersensitive teeth are generally not good candidates for bleaching, although management of the sensitivity may allow successful bleaching of their teeth. Because bleaching tends to produce some sensitivity under ordinary circumstances, patients with preexisting tooth sensitivity must be cautioned that increased sensitivity, albeit transitory, will occur.[53] However, there are reasonable techniques, discussed later, to manage sensitivity.

Box 16-2	**Guarded prognoses for bleaching**

- History of sensitive teeth
- Translucent teeth
- Extremely dark gingival third of tooth that is visible when smiling
- Exposed root surfaces
- Extensive white spots that are very visible
- Temporomandibular disorder (TMD) or bruxism

Most teeth become more opaque upon bleaching, but some that are already translucent at the incisal edge may become more translucent and may not appear to whiten in that area (Fig 16-4). Patients should be informed of this potential challenge. One treatment for this situation, if the occlusal examination indicates it is possible, is the placement of bonded composite restorations on the lingual tooth surface to reduce the translucency. Teeth heavily stained in the gingival third have the poorest prognosis for complete lightening, as the root dentin and the deep dentin of the tooth are very different from dentin near the dentinoenamel junction in the shape, size, and number of dentin tubules per unit area.[53] Roots of teeth do not bleach as well as anatomical crowns, so patients with gingival recession that exposes roots will display yellow tooth structure in those areas.[54] White spots do not bleach but may be less noticeable if the background color of the tooth is lightened to avoid the contrast. Patients with a history of temporomandibular disorders (TMDs) may not be good candidates for at-home bleaching or may need to wear the tray during waking hours only. A special tray design that covers only the facial surfaces of the teeth may be helpful.[55] Bruxers also may have to alter wear times for at-home treatment or have several trays fabricated during treatment because they will often grind holes in the tray.



Fig 16-4 *(a)* A translucent incisal edge can often appear discolored and should be tested for discoloration. Prior to bleaching, the incisal edge appears bluish because of the darkness of the mouth seen through the translucent enamel. *(b)* To test for discoloration, block the light transmission by placing a white-gloved finger on the lingual of the tooth. If the discoloration disappears, the incisal edge is translucent; if it remains, the incisal edge is discolored. Discoloration responds well to bleaching, but translucency may create an esthetic problem. *(c)* A translucent incisal edge often becomes more translucent after bleaching. Patients should be informed of this possibility prior to bleaching. *(d)* After bleaching, if the translucency is unacceptable to the patient, some mechanism is needed to block the light transmission. The potential for improvement is evident by using the finger test again. Often a lingually placed bonded composite restoration can minimize translucence, as long as the restoration does not interfere with the patient's occlusion.

There is some evidence to suggest that the use of carbamide peroxide may retard the progression of caries.[56–61] Therefore, in carefully selected cases it is not essential to restore all active caries lesions prior to bleaching. Because it cannot be predicted to what shade a tooth will bleach, there is no way to know the proper shade to select for the restoration. It is best to perform bleaching and then select the shade after the proper stabilization time has passed. Only those caries lesions that are sensitive or near the pulp would need to be restored prior to bleaching and then possibly later resurfaced or replaced after bleaching to match the final shade.[62,63] Bleaching does interfere with the bonding process because it results in a very high oxygen concentration in the enamel and dentin, which hinders polymerization of the resin composite.[64–66] A 25% to 50% reduction in bond strength of composite to etched enamel, due to oxygen inhibition of polymerization of the adhesive resin tags within the etched enamel, has been reported. Some research has found that a delay of 7 to 10 days after bleaching allows dissipation of the excess oxygen from the tooth structure so that there is no interference with the polymerization reaction.[67] Another study showed that there is a need to wait at least 2 weeks for maximum bond strength of silorane-based or methacrylate-based restorations; otherwise, the enamel bond-strength reduction is 25% to 50%.[68] Waiting 1 to 2 weeks is also important in resin bonding to allow the shade of the bleached teeth to stabilize.[69] Patients with existing esthetic restorations must be warned that when bleaching lightens the natural tooth color, restorations may appear relatively dark and unattractive. The need for new restorations that are lighter in shade should be discussed with the patient prior to bleaching. Occasionally, the cost and number of restorations that would require replacement after bleaching or the danger of replacing a restoration such as a crown on a cracked tooth may be a contraindication to bleaching.

Mode of Action

The bleaching process is designed to enable the oxidizing agent to reach sites within the enamel and dentin to allow a chemical reaction to occur. No matter the bleaching technique or specific bleaching action, the intention is to deliver the active ingredient to the discolored segments of the tooth to dislodge or decolor the chromatic particles.

Hydrogen peroxide diffuses through the organic matrix of the enamel and dentin because of its low molecular weight.[70–74] It is not currently understood what gives the tooth its genetic color, and therefore it is not possible to state scientifically what exactly the bleaching process is changing within the tooth. Bleaching both removes extrinsic and intrinsic stains in or on the tooth that originated from external sources (nicotine, tetracycline, coffee, etc) as well as changes the genetic color of the tooth. People are born with different colors of teeth in the same manner as they have different colored eyes and hair. The teeth will change color at different rates for different patients, so the dentist cannot predict how long bleaching will take or what the final shade will be. Moreover, there is a limit to how white a tooth will become, and this limit differs from patient to patient. Further research is needed to determine what gives the tooth its baseline color and, in turn, the extent of the effect of the oxidation reaction that changes the tooth color during bleaching.

The chemistry of carbamide peroxide used in at-home bleaching is thought to be a bit different from that of hydrogen peroxide, although the final stages do involve the reaction of hydrogen peroxide with the compounds within the tooth. When introduced into the mouth, a solution of 10% carbamide peroxide agent breaks down into 6.5% urea and 3.5% hydrogen peroxide, both of which access the internal portions of the tooth in minutes (Fig 16-5). This 3:1 reduction of carbamide peroxide to the active hydrogen peroxide ion explains why bleaching formulations with carbamide peroxide typically are dosed at three times the concentration of those containing hydrogen peroxide. Bleaching not only removes discoloration from within the tooth, but it also alters/brightens the inherent color of the dentin itself.[75]

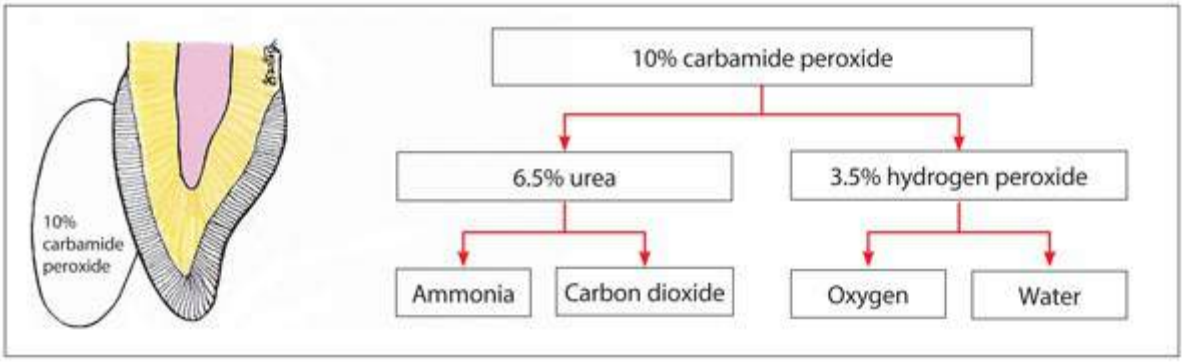


Fig 16-5 Carbamide peroxide penetrates through the enamel and the dentin to the pulp in 5 to 15 minutes. Carbamide peroxide dissociates in the tooth into urea and hydrogen peroxide. These two components further break down into ingredients that are common in the body and are easily metabolized by the body. Urea serves to elevate the pH of the local environment.

Treatment Planning and Patient Education

A basic understanding of the cause(s) of discoloration is necessary to better predict the course and duration of the treatment as well as the final outcome. The decision to use at-home or in-office bleaching is based on the patient's preference, financial situation, and ability and willingness to comply with the treatment protocol. Patients who are unwilling or unable to comply with the protocol for the at-home technique, or who are eager to finish bleaching in a very short period of time regardless of cost, are good candidates for in-office bleaching. Subsequent treatment procedures should be planned so that limits of the bleaching treatment are discussed in the context of solving the patient's other dental problems. The patient should be informed that the shade of any restoration placed previously will not be altered by bleaching, so bleaching should be performed before any esthetic restorative procedures. Information concerning the decision to bleach or not, as well as the rationale for and costs of choosing a particular method, must be recorded to verify that the dentist and patient agree on the procedures and their predicted outcomes.

A proper examination is essential to diagnosing the cause of discoloration as well as to determine if there are other contributing factors or concerns to address[76] (Box 16-3). Discoloration or staining may be an indication of dental conditions such as a necrotic pulp (Fig 16-6), decay, resorption (Fig 16-7), or faulty restorations. Without a proper examination and diagnosis of the cause of discoloration or staining, the disease process occurring in the tooth may be masked by removal of the only clinical symptom of the problem. For example, dark teeth could be a result of trauma to the tooth at an early age, and the tooth pulp may eventually become necrotic or abscess. It may take 1 to 20 years before this condition occurs.[77] Occasionally,

external resorption may be found clinically when carefully exploring at or beneath the gingival tissue, as this process generally occurs on the root of the tooth. However, both internal resorption and external resorption often are only found through radiographic examination. Bleaching discolored teeth for which the color change is the only indication of hidden pathology will change the color of the tooth while not resolving the pathology that is causing the discoloration. With the possible exception of caries, bleaching does not stop the pathology from continuing, and allowing the pathology to continue can result in the loss of teeth. Additionally, caries alone or around restorations may cause teeth to appear dark. A dental examination, complete with a screening radiograph of the anterior teeth and any single dark teeth, is recommended prior to initiating any type of bleaching. The examination should include an explanation of all treatment options available to the patient, taking into consideration the fact that existing restorations will not bleach, as well as other esthetic needs. After a proper examination, the cause of discoloration is diagnosed and the treatment plan developed (Box 16-4). Correctly diagnosed indications for bleaching may require varying treatment times or techniques, based on the type and depth of the stain/discoloration and the number of teeth involved. Single dark teeth may require a special single-tooth tray or multiple tray designs.[78]



Fig 16-6 A screening radiograph should always be taken of any anterior teeth being considered for bleaching, including a single dark tooth. *(a)* A left mandibular lateral incisor that is darker than adjacent teeth. This tooth was asymptomatic. *(b)* Periapical radiograph reveals apical radiolucency associated with the darkened tooth.



Fig 16-7 Root resorption without bleaching. The only indication of a problem with this tooth was a slight discoloration. The lesion was discovered on clinical and radiographic examination. (Courtesy of Tom McDonald, Athens, Georgia.)

Box 16-3	**Bleaching examination considerations**
• Intraoral soft and hard tissue examination (to rule out cancer, periodontal problems, recession, etc) • Radiographs (to assess periapical pathology, resorption, and single dark teeth) • Diagnosis of cause and location of tooth discoloration • Identification of esthetic restorations • Assessment of exposed roots, visible white spots, cracks, tooth or gingival architecture defects; overall esthetic/facial analysis; other appliances; and lifestyle issues • Complete sensitivity history and examination • Occlusal and TMD screening examinations	

Box 16-4	**Diagnosis of tooth discoloration**
Generalized tooth discoloration	**Single-tooth discoloration**
• Genetic color of the teeth	• Endodontic therapy
•Tetracycline staining	• Necrotic pulp with periapical radiolucency
• Coffee, tea, nicotine use	• Internal or external resorption
• Multiple caries	• Calcific metamorphosis
• Multiple discolored restorations	• Silver point endodontic obturation or restorative material discoloration
• Multiple leaking restorations	• Caries
	• Discolored restoration
	• Leaking restorations

In addition to diagnosis of the cause of the discoloration, the dentist also should consider the overall (Box 16-5) and esthetic conditions (Box 16-6) of the patient. Certain changes in the treatment protocol may be required based on these

considerations.

Box 16-5	**Overall considerations for bleaching**

• *Sensitivity:* History of sensitivity and plans for sensitivity treatment before or during bleaching
• *Tray design:* Tray design for full-arch or single-tooth tray; decision to extend the tray onto tissue or to have no tissue con
tact; decision to incorporate spacers on teeth or to include other design features, such as no occlusal coverage for TMD
patients
• *Treatment time:* Daytime or nighttime usage plans for tray bleaching
• *Appliances:* Other appliances being worn with tray bleaching, such as orthodontic appliances or removable partial dentures
• *Combination treatment:* In-office bleaching requiring multiple visits versus a combination of in-office treatment followed by
tray treatment versus tray only treatment
• *Joint health:* TMD that may be aggravated by tray wear or by extended opening during in-office bleaching
• *Pregnancy:* Decision to not perform bleaching procedures on pregnant or nursing mothers for the psychologic concerns
related to birth issues, as well as pregnancy gingivitis and morning sickness

Box 16-6	**Esthetic considerations for bleaching**

• Compare the color of the teeth with the color of the sclera of the eyes to ascertain the whitening outcome potential.
• Determine if the patient has a ″gummy smile“ and short teeth, because whiter teeth accentuate excessive gingival display.
Periodontal therapy may be indicated.
• Identify other restorations that are displayed in a full smile, because they will not change color.
• Note alterations in individual tooth gingival architecture that may be more noticeable after bleaching.
• Point out exposed root surfaces that will not bleach, *and* demonstrate white spots that will not change color.
• Evaluate transparent areas of the incisal edge that may become more noticeable *and* discolored gingival areas that are
most difficult to bleach, especially tetracycline-stained areas.
• Identify dark restorations on the lingual or internal aspect of endodontically treated teeth that may alter the effectiveness of the color change.

Shade selection and record collection

A shade guide identifies the existing tooth color to establish the baseline (Fig 16-8). If the shade guide does not have a match for the tooth color, it should be estimated. It is important that the patient agrees that the shade tab is the closest match to the

current shade of the teeth. This shade should be recorded in the chart with the patient observing the entry. It is also recommended to make preoperative photographs with the shade tab in the photograph. The advent of cost-effective digital dental cameras has made this mode an excellent method of recording the initial, interim, and final treatment conditions of the patient.



Fig 16-8 The shade guide tab establishes a record of the existing color. Digital photographs should be taken with the tab visible and the teeth edges in close approximation.

If using at-home bleaching techniques, the best way to ensure patient awareness of the effect of bleaching is to bleach one arch at a time. In addition to minimizing the potential for temporomandibular joint (TMJ) problems, it preserves one arch for an ongoing comparison of progress. Another reason for treating one arch at a time is to enable determination of the effectiveness of the bleaching procedure before the patient commits to bleaching of the second arch. If the first arch is deemed a success by the patient, the opposing arch can be bleached for an additional fee (Fig 16-9). Finally, bleaching one arch at a time reduces the number of teeth for which sensitivity may be a problem. The American Dental Association (ADA) reimbursement code for tray bleaching is also per arch, not per patient.



Fig 16-9 Note the strong contrast between the bleached maxillary teeth and the unbleached mandibular teeth (which match the shade tab). Bleaching one arch at a time is preferred to allow the patient to see that the process is working and to minimize sensitivity and occlusion issues. Some patients may choose not to bleach the opposing arch.

Impressions for diagnostic casts are necessary if bleaching trays are to be used for the at-home technique. The impressions can be made with alginate impression material if immediate pouring is planned. For delayed pouring, a polyvinyl siloxane impression from a putty-wash technique is more appropriate. Patients who gag easily may prefer an impression with a double-sided full-arch tray using a closed-mouth technique or use of a mandibular tray on the maxillary arch, or they may be candidates for a thermoplastic tray fabricated directly in the mouth without impressions.[79]

For added documentation, close-up photographs of the patient's teeth and a full-face photograph of the patient with a full smile should be kept as part of the patient's record. The photographs may be helpful in treatment planning, in reminding the patient of the original appearance of the teeth, and as a record for a ceramist fabricating any restorations. Photographs should be taken at a consistent magnification and pose. Placing the incisal edges of the teeth in the bleached arch in close proximity to the teeth in the unbleached arch is important so that both arches are the same distance from the camera; this helps to ensure the best comparison of the before and after shades, emphasizing the effect of bleaching.

Patient education

The patient should be well informed about the bleaching procedure. Information explaining the process, precautions, possible side effects, number of applications or appointments, anticipated total time for bleaching, and likely results should be provided. As with any patient instructions, providing written information concerning the procedure as well as treatment instructions is recommended. The dentist should also explain why the particular technique has been chosen. The steps of the procedure and consequences for not following them should be outlined. For the at-home technique, loading and insertion of the bleaching tray should be demonstrated and the length for each wear session outlined. Any questions the patient might have should be answered at this appointment. The patient should sign a consent form indicating that he or she has been informed about the procedure, its expected outcome, and any potential side effects. The consent form should also list other treatment options for this condition and state that those remain if the bleaching is not successful.

Current Bleaching Modalities

The three types of bleaching therapy are in-office bleaching, at-home bleaching, and over-the-counter products. In-office bleaching is the oldest method and may involve heat and light enhancement. At-home bleaching involves the application of a lower concentration of peroxide applied at home in a tray fabricated at the dental office. The third type of bleaching involves over-the-counter (OTC) products, which are totally consumer-managed without any examination, guidance, or instructions from the dentist. Generally, bleaching can first be categorized into treatment for either endodontically treated teeth or vital teeth. Furthermore, endodontically treated teeth can be bleached in the office, outside the office, or a combination of both inside and outside the office. Outside-the-office (at-home) treatment consists of applying internally as well as externally to the tooth a material that actively lightens it while the patient is away from the office. The in-office technique accomplishes the application of the bleaching material in the office. Vital tooth bleaching also offers a choice between an in-office technique and an at-home technique. In-office techniques include the application of a bleaching material to teeth isolated by rubber dam and may include enhancement of the process by heat or light. At-home bleaching can be dentist-supervised or patient-managed. Dentist-supervised at-home bleaching uses a lower concentration of peroxide than in-office bleaching and may use carbamide or hydrogen peroxide, which is applied in a custom-fitted tray that the patient wears at home, usually while sleeping. The consumer-managed at-home bleaching systems are available in the form of tray systems, trayless systems, and paint-on products.

Nonvital tooth bleaching

Endodontically treated teeth are especially susceptible to discoloration from blood products caused by trauma or endodontic therapy or by necrotic tissue inadvertently left in the pulp chamber. Endodontically treated teeth may be treated internally, externally, or both (Fig 16-10).

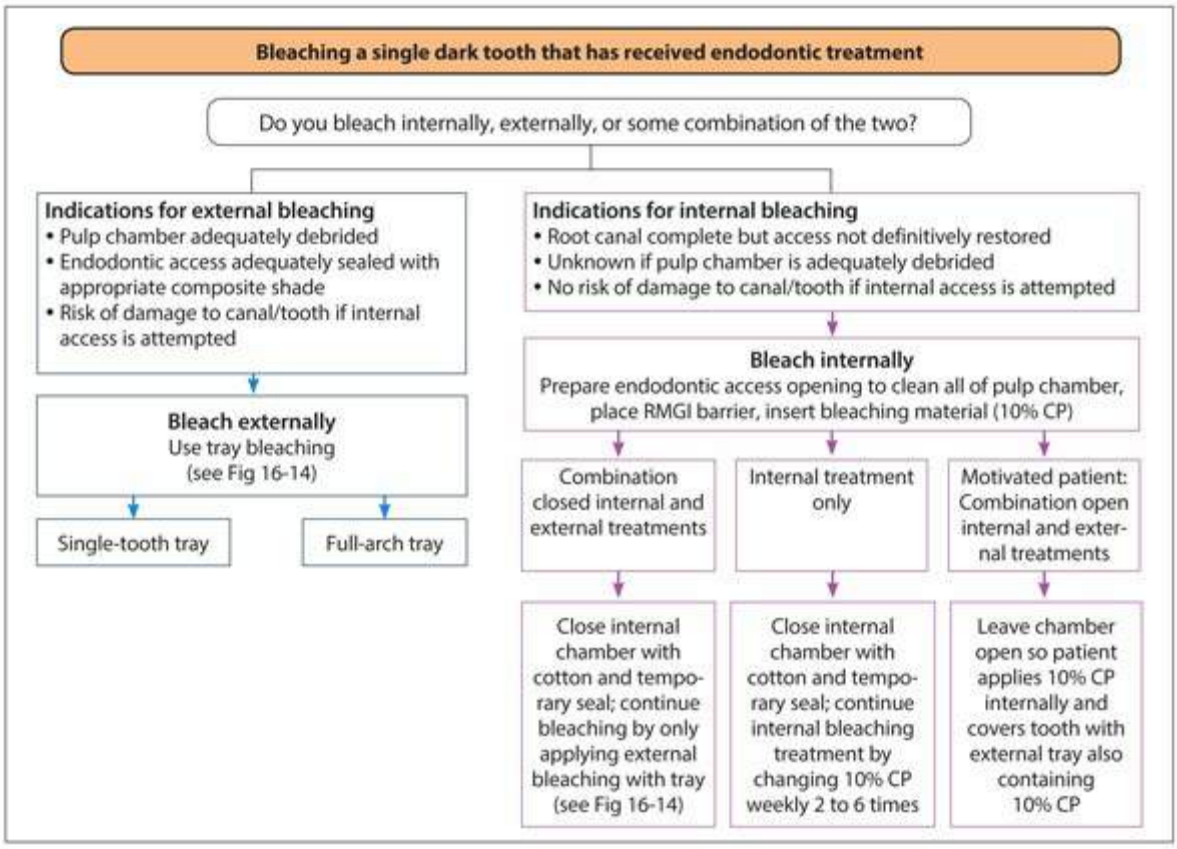


Fig 16-10 Algorithm for bleaching options on a single dark tooth that has received or needs endodontic therapy. These teeth can be bleached internally, externally, or with some combination of the two techniques. RMGI—resin-modified glass ionomer; CP—carbamide peroxide.

Internal bleaching

For a nonvital tooth that has received endodontic therapy, internal bleaching is a good option if the root canal treatment is complete but the endodontic access is not restored. The restorative dentist should open the access opening sufficiently to reach the incisal and lateral extents of the pulp chamber and clean it out (Fig 16-11). After the pulp chamber is cleaned, the gutta-percha is removed 2 mm below the cementoenamel junction, resin-modified glass ionomer is placed over the gutta-percha, and the bleaching agent is applied.

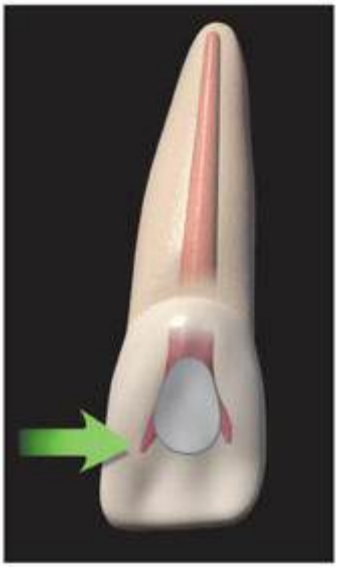


Fig 16-11 An endodontic access opening may not provide access to pulp tissue in the pulp horns *(arrow)*, which can cause darkening of the tooth. This is especially common in young teeth with large pulps. Access should be made to remove the remaining tissue. (Courtesy of Gary Chike, Augusta, Georgia.)

A historical bleaching method dating to the 1800s used 30% hydrogen peroxide applied to the pulp chamber in one of two techniques (Fig 16-12). One was a total in-office technique known as the *thermocatalytic technique*, and it involved the use of heat applied several times during a 30-minute period to enhance the action of the solution in the pulp chamber, after which the solution was rinsed out. The alternative to a total in-office procedure, called a *walking bleach*, used a paste of watery 30% hydrogen peroxide and sodium perborate powder that was sealed into the chamber to permit activation of the solution over several days. The patient returned weekly, and the solution was changed one to four times until the maximum lightening of the tooth was achieved. The two techniques were equally effective.[80] Internal bleaching by either method has been shown to return teeth to their desired color in 83% to 91% of cases.[81,82] However, because of the caustic nature of the high concentration of hydrogen peroxide and the possible detrimental effect of resorption, techniques using high concentrations of hydrogen peroxide are no longer recommended. Other safer options for walking bleach include the use of sodium perborate mixed with distilled water or anesthetic, or 10% carbamide peroxide sealed in the pulp chamber. The results of an in vitro study[83] demonstrated that after three applications there was no difference in the efficacy of sodium perborate mixed with water and sodium perborate mixed with either 5% or 30% hydrogen peroxide. Another study[84] showed no root resorption after 3 years using sodium perborate, with a 90% esthetic success rate initially and a 49% esthetic success rate after 3 years. A newer technique

involves the placement of 10% carbamide peroxide sealed into the chamber. This lower concentration of peroxide also provides a large safety margin. Additionally, one of the benefits of carbamide peroxide is the elevation of the pH, which may combat resorption. For the best combination of safety and effectiveness, the recommended treatment for internal bleaching is sodium perborate mixed with water, or 10% carbamide peroxide alone, used after the placement of a protective barrier material in the cervical area.

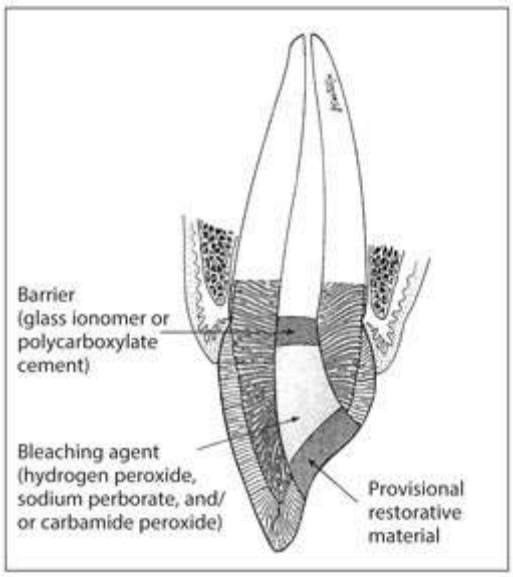


Fig 16-12 The restorative material blocking the canal at the cementoenamel junction seals the endodontic root filling material from the chamber and decreases dispersion of the bleaching agent through the root dentin.

A more serious problem than the danger of tissue burns from 30% hydrogen peroxide is the occurrence of external root resorption (Fig 16-13). This problem affects about 7% of the teeth that have undergone internal in-office bleaching[85–87] and often causes tooth loss.[85,88–90] While the causes of root resorption are not fully known, a review of the literature indicates a number of possibilities. The patients in which root resorption occurred tended to be younger than 25 years old, and most had a history of traumatic injury. Some underwent bleaching with the application of heat and some did not, but heat does appear to be a causative factor. Animal studies have shown a cause-and-effect relationship between internal bleaching with 30% hydrogen peroxide and resorption, with a resorption incidence of 18% to 25% when heat was applied[91,92] and 0% to 6% without heat application.[91,93] Any one of several factors may also need to be present for resorption to occur, including: *(1)* deficiency in the cementum, exposing the cervical dentin to the oral cavity (normally affecting approximately 10% of the population); *(2)* injury to the periodontal ligament,

triggering an inflammatory response (trauma); *(3)* infection, sustaining the inflammation; *(4)* lack of a seal over the gutta-percha; *(5)* high heat; and *(6)* high concentration of hydrogen peroxide.



Fig 16-13 Severe external root resorption in the lateral incisor. This situation may result with or without bleaching and is found by clinical and radiographic examination. (Courtesy of Sandra Madison, Asheville, North Carolina.)

Cementum deficiencies expose permeable dentin that can allow toxic substances and bacteria from within the chamber and root canals to emerge at the root surface. This may cause an inflammatory process in the periodontal ligament.[94] A 30% solution of hydrogen peroxide is caustic enough to alter the chemical structure of cementum and dentin,[95] decrease their microhardness,[96] increase resorption,[97] and enhance transtubular movement of bacteria.[98] The solution diffuses through the radicular dentinal tubules at an increased rate if cementum deficiencies are present[90] and when heat is applied.[99] These data indicate that internal bleaching with 30% hydrogen peroxide is not as safe as originally believed. However, in addition to cementum defects, a history of trauma and marked overheating are major factors in resorption.

Research has been conducted to determine if a protective restorative material can be placed in the cervical portion of the tooth to prevent this resorption problem. Unfortunately, this barrier layer reduces the diffusion of hydrogen but does not prevent it[92,100] and does not necessarily protect the tooth against root resorption.[101] However, most of the teeth reported with cervical resorption did not have a barrier placed over the gutta-percha. It is not prudent to use heat or high concentrations of peroxide, and a barrier layer should be placed over the gutta-percha (Box 16-7).

Box 16-7	**Recommendations for internal bleaching of a single dark nonvital tooth to avoid resorption**

•Use a low concentration of peroxide (10% carbamide peroxide or sodium perborate); 10% carbamide peroxide has the potential to increase the pH to basic, which may also prevent resorption.
• Place a base over the root-filled portion of the tooth (glass ionomer or resin-modified glass ionomer).
• Do not use high concentrations of peroxide, high heat, or light activation.
• Consider a single-tooth bleaching tray for external bleaching once the internal portion is cleaned and prepared or later for touch-up if there is a color relapse after restoration.

The effect brought about by internal bleaching is generally not long-term. Within 1 to 5 years, only 35% to 50% of the teeth maintain their esthetically pleasing appearance.[81,82,85] Therefore, the process must be repeated periodically. Unfortunately, rebleaching presents a problem for a tooth in which the access preparation was restored with resin composite. Attempting to rebleach internally would require removal of the previously bonded resin composite restoration. This removal generally results in additional loss of tooth structure, which weakens the tooth. A better treatment option once the tooth has been bleached internally and the lingual access restored with resin composite is to bleach the tooth externally using the same techniques used for vital teeth. This can involve the use of either a single-tooth bleaching tray or a conventional tray covering all the teeth.

External bleaching

Tray bleaching works well on vital (Fig 16-14) and nonvital single dark teeth. The choice for bleaching can involve a tray covering all the teeth, as described later, or a single-tooth bleaching tray. A single-tooth tray is fabricated by first making the traditional tray, then removing the part of the tray covering the teeth on either side of the dark tooth (Fig 16-15). The patient applies the carbamide peroxide externally in the single dark-tooth mold and wipes the excess from the adjacent teeth. Use of the single-tooth tray allows the dentist to determine the maximum whitening of the darkest tooth prior to changing the color of the adjacent teeth. If the darker tooth does not lighten as much as the other teeth, then no treatment is applied to the adjacent teeth. If the single tooth gets lighter than the adjacent teeth, then a full tray can be used sparingly to match the adjacent teeth to the lighter single tooth. To avoid overlightening the adjacent teeth beyond the maximum lightening achieved by the original single dark tooth, hourly daytime wear rather than overnight treatment is

advocated.

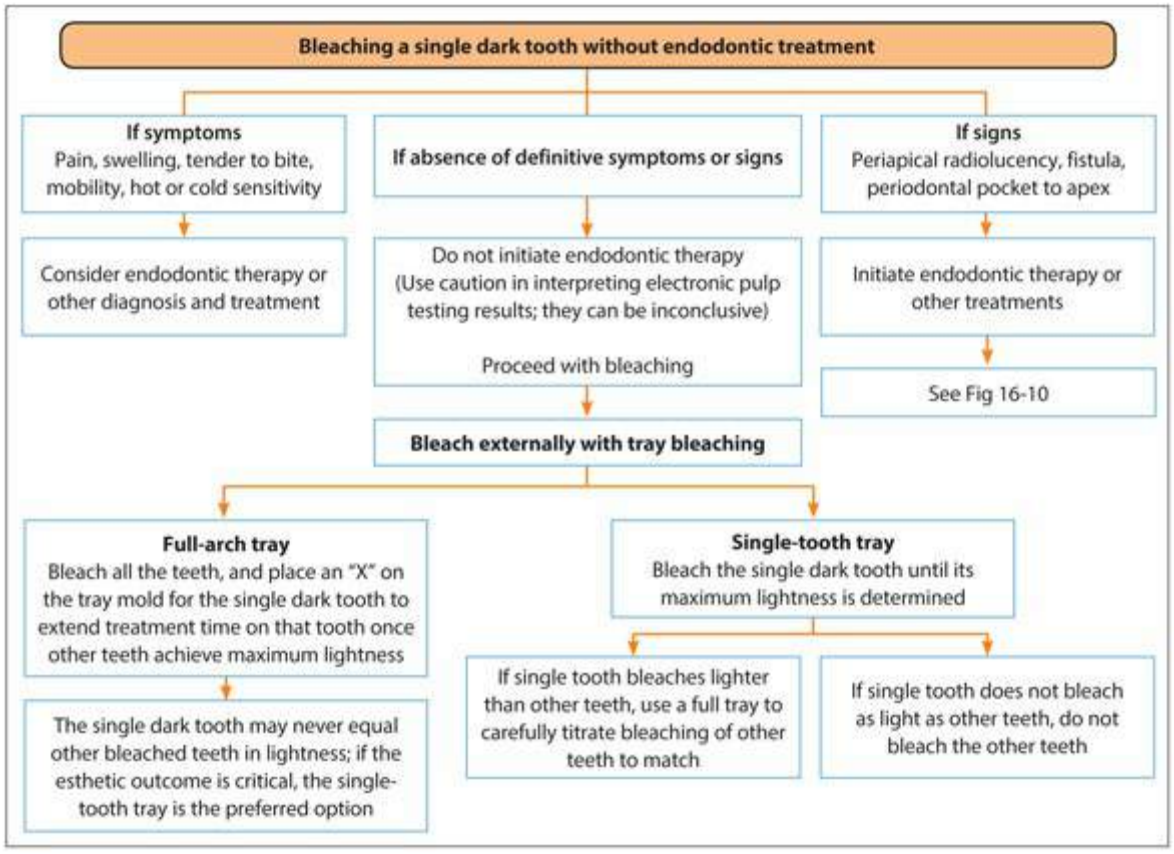


Fig 16-14 Algorithm for bleaching a single dark tooth that has not received or does not require endodontic therapy.

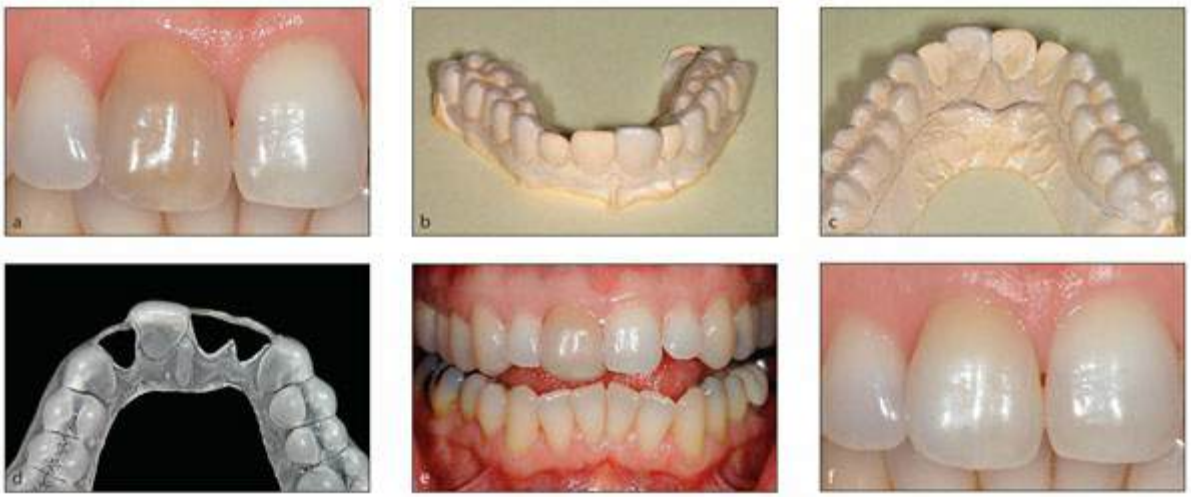


Fig 16-15*(a)* The best treatment for a single dark tooth is to fabricate a single-tooth bleaching tray. This tray allows the dentist to determine the maximum lightening of the single dark tooth prior to initiating any bleaching on the other teeth. *(b and c)* The single-tooth bleaching tray has to extend further onto the palate for retention, because the tray covering the teeth on either side of the single dark tooth will be removed to avoid bleaching them. *(d)* The tray is trimmed to fit around only the discolored tooth. *(e)* The patient places bleaching material only in the single dark tooth mold and inserts the tray. Excess gel extruding onto the adjacent teeth is wiped away by the

patient. *(f)* The discolored tooth is whitened while other teeth are not affected.

Internal-external bleaching, open and closed

Another technique, which has limited application, is the *internal-external open bleaching technique*. It is accomplished by preparing the tooth in the same way as you would for the walking bleach technique combined with the at-home tray technique. That is, the pulp chamber is left open to allow the patient to place 10% carbamide peroxide inside the chamber and, at the same time, apply it externally with the tray. This technique, called *internal-external bleaching*, is very effective and may reduce the treatment time by as much as 50%.[102,103] However, it is best suited for patients who are very responsible and capable of applying the solution intraorally. With this technique, a barrier layer should be placed over the gutta-percha to prevent contamination of the root canal, but concern exists that the patient may not return to have the endodontic access opening restored, resulting in caries and loss of the tooth from neglect.

A more practical application of this combined technique is the *internal-external closed bleaching technique*, where the dentist prepares the tooth for the traditional walking bleach technique and seals the chamber containing the 10% carbamide peroxide. However, rather than return for multiple changes of the peroxide paste, the fabrication of a single-tooth bleaching tray[77] allows the patient to continue the process of bleaching at home until the tooth reaches its maximum lightness. This approach allows for better patient and office management, as some single dark teeth may require 6 to 8 weeks of bleaching to resolve the discoloration. This same tray can be used for touch-up of the single tooth should it darken after several years. If the single tooth gets lighter than the adjacent teeth, then the original cast can be used to make a full tray. The full tray can then be used very sparingly during the day to lighten the adjacent teeth to match the new final shade of the single dark tooth. Conversely, if the dark tooth does not become as light as the adjacent teeth, then the adjacent teeth will remain untouched for the best esthetic outcome.

Once the tooth has reached its maximum lightness, the closure of the access opening is delayed for 2 weeks to allow the shade to stabilize and the bond strengths to return to normal values. At that time, the final shade is compared with the adjacent teeth to determine if any restorative adjustments are needed to match the single dark tooth to the adjacent teeth. Decisions must be made as to whether to place a stark white resin composite in the chamber to give a lighter appearance or to match the resin composite shade to the tooth shade exactly and whether to use a translucent or opaque composite restoration for closure (Fig 16-16).

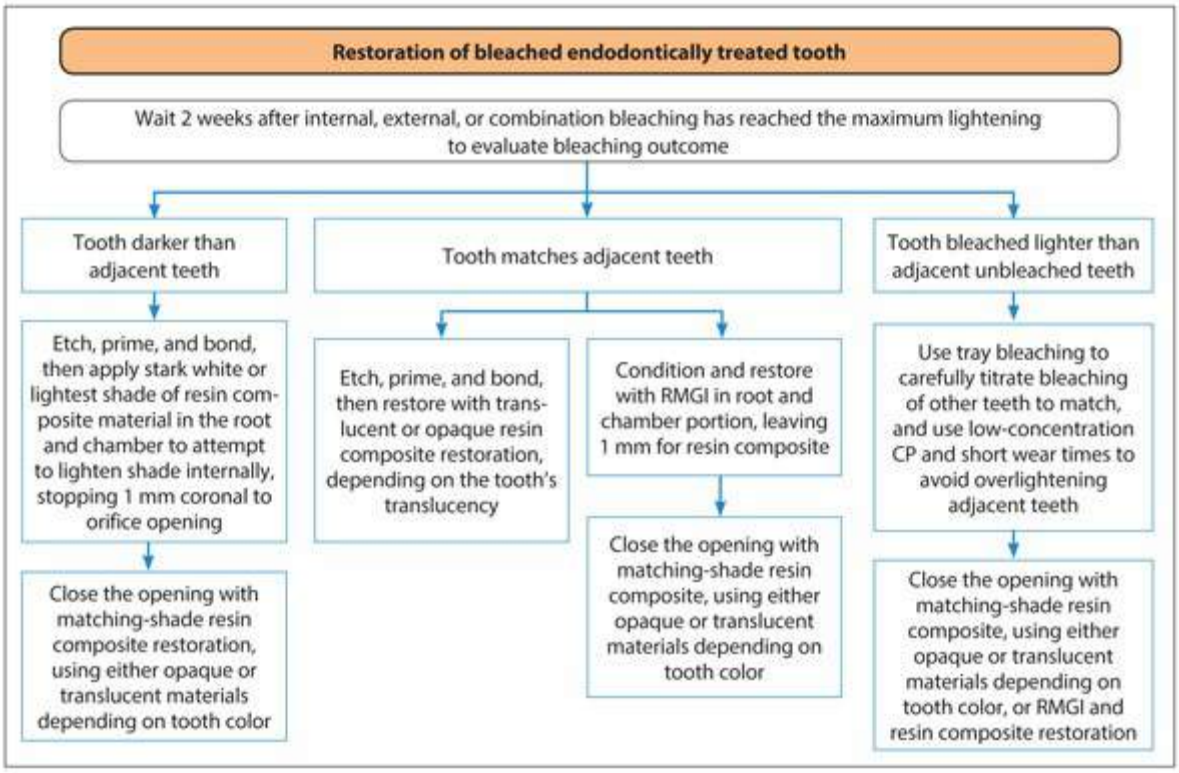


Fig 16-16 Algorithm for restoration of a bleached endodontically treated tooth. RMGI—resin-modified glass ionomer; CP—carbamide perioxide.

Vital tooth bleaching

Vital bleaching may be accomplished by either the in-office technique or take-home trays, and each has some variations. The in-office technique currently is not as popular as the at-home tray technique, but it has historically met with success, and there are some specific indications for its use.

In-office technique

In-office bleaching of vital teeth generally uses a 35% hydrogen peroxide solution placed directly on the teeth and may involve application of light and/or heat to enhance the peroxide release.[15,104] Because the hydrogen peroxide concentration is so high and the material is so caustic, soft tissues must be very well protected to prevent injury[74] (Fig 16-17).This technique is intended to produce the bleaching effect with limited need for patient compliance. It is indicated for achieving more rapid results (Fig 16-18) or for patients who may have difficulty following the regimen for the at-home technique (Box 16-8). Several questions must be asked (Table 16-1), and the potential disadvantages need to be taken into consideration: The fee is usually higher because more chair time is required; there is a possibility

of tissue injury from the more potent agent used (Fig 16-19); and the results of one in-office treatment may be inferior to the slower at-home method. In-office bleaching may require more than one visit to achieve the desired results.[105] In fact, manufacturers of in-office bleaching systems usually recommend at-home bleaching after in-office treatment is completed.[106] One study found that in-office bleaching combined with at-home bleaching does not improve tooth whitening compared with at-home bleaching alone.[107]

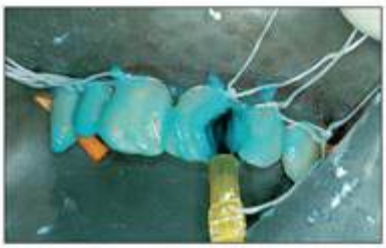


Fig 16-17 Rubber dam protects the soft tissues during an in-office bleaching procedure.



Fig 16-18 In-office bleaching can produce good results quickly but takes an average of three appointments. *(a)* Prior to bleaching. *(b)* After one bleaching session. (Courtesy of Sandra Madison, Asheville, North Carolina.)



Fig 16-19 A high concentration of hydrogen peroxide can produce a chemical burn, as shown on this patient's lip.

Box 16-8	**Indications for in-office bleaching**
• Patient is fully informed of all bleaching options before making the treatment choice. • Patient has financial resources to continue treatment beyond one appointment and for additional tray treatment should that be indicated to complete the initial bleaching.	

• Patient is unwilling or unable to use at-home tray application.
• Patient desires the process to be completed as quickly as possible due to an upcoming event.
• Patient does not have sensitive teeth or will premedicate with analgesics for sensitivity as needed.
• Patient does not want posterior teeth as light as anterior teeth, or posterior teeth do not show in smile analysis.
• Patient is willing to pay again for touch-up, office, or tray treatment in later years.
• Patient needs encouragement for compliance with tray bleaching.
• Patient understands that the outcome is the same with all bleaching techniques if treatment continues long enough;
cost-benefit and risk-benefit ratios are considered.

Table 16-1	**Questions regarding in-office bleaching**
Question	**Answers**
Does one in-office bleaching treatment yield the same outcome as tray bleaching?	Typically, no: The average is three in-office sessions to reach maximum tooth white ness, while the range is one to six treatm
Do the lights make a difference in the outcome of in-office bleaching?	No: Dehydration may give initial lighter appearance; tooth color changes at a certain rate regardless of "enhancement" of peroxide by the light.
When is the best time to evaluate the color change from in-office bleaching?	Two weeks or longer after treatment due to dehydration from isolation and heat (if used).
Is a combination of in-office and tray bleaching the best option?	Only if the patient wants to pay extra for the initial boost: The final outcome is the same whether one or the other or a combination is used.
What are the sensitivity issues for in-office bleaching?	Sensitivity with in-office treatment is greater than with tray bleaching, so sensitivity must be minimized by using shorter appointments on multiple visits rather than a single long appointment; patients may need to premedicate with nonsteroidal anti inflammatory drugs.

With in-office bleaching, a portion of the whitening effect is temporary, resulting from dehydration of teeth from the isolation technique. Dehydration alone has been shown to cause a transient 3- to 12-shade change (on a VITA Classic shade guide), or a ΔE (using a colorimeter) of 6.7 (which is approximately the difference between B1 and D3 or between A2 and B4 on the VITA Classic shade guide) [108] after 1 hour.[109] The effects of bleaching using rubber dam isolation must be evaluated after

1 to 2 weeks to allow rehydration of the teeth to a stable shade. Authors suggest from one to six in-office treatments, with the average being about three visits.[105] The lighter the teeth at the start of bleaching, the fewer visits required.

Because in-office bleaching utilizes a much more potent agent, the oral/perioral structures must be protected during the procedure. Generally, this is best accomplished with a well-placed rubber dam tightly adapted around the cervical areas of the teeth.[110,111] For the single isolated tooth or a minimal number of teeth, it may be possible to protect the gingiva with cotton rolls and a so-called liquid rubber dam (light-polymerized resin) (Fig 16-20). The light-polymerized resin is injected onto the gingival tissues surrounding the target teeth and then light cured to form a flexible protective shield that can be removed easily. Any material covering the teeth should be trimmed to the proper extension to allow the bleach to reach that area. Cotton rolls and cheek retractors may be needed to protect the lips and tongue.



Fig 16-20 A resin "paint-on" dam is applied and light cured around the target teeth. A 35% hydrogen peroxide bleaching material has been applied.

The teeth should not be anesthetized for in-office bleaching. This allows the patient to note any developing discomfort so the dentist can avoid overtreating the teeth during the several applications of the bleaching solution occurring in the appointment. The patient becomes the "control" for the number of applications that should be placed; he or she should be questioned after each application about any tooth discomfort or tingling in the gingiva, which likely indicates a gap forming between the teeth and the rubber dam. If any tooth discomfort is experienced, the treatment should be interrupted and continued at a future appointment to allow the teeth to recover. If the patient reports a tingling of the gingival tissues, the procedure should be immediately aborted, the dam removed, and the tissues rinsed with water (with or without baking soda) to neutralize the peroxide and avoid severe tissue burns (Fig 16-21). Although there are no clinical studies on treating tissue burns from bleaching solutions, empirical reports have suggested that in addition to water with baking soda, calcium hydroxide powder mixed with distilled water (Alejandro Iglesias, personal communication, 2010), as well as a zinc oxide–eugenol provisional cement (TempBond, Kerr) (T. Bob Davis, personal communication,

2011) are effective (Fig 16-22).



Fig 16-21 The gingival tissue around the canine has been chemically burned by contact with 35% hydrogen peroxide.



Fig 16-22*(a)* A soft tissue burn from a mild peroxide solution, resulting in a painful experience for the patient and white discoloration of the tissue. *(b)* Application of a zinc oxide–eugenol provisional material (TempBond) for 2 to 3 minutes until pain subsides. (Technique courtesy of T. Bob Davis, Dallas, Texas.) *(c)* Once the zinc oxide–eugenol provisional material is removed, both the whitish area and the pain are gone.

Manufacturers provide their bleaching agents in various forms. Some are very watery, while others are in a gel form. Some are packaged as a powder-liquid combination that is mixed to activate and then placed on the teeth, while others are provided as a ready-to-place solution in a syringe. The solution is directly applied to the facial and proximal areas of the teeth for the prescribed time interval. If access is limited because of the presence of restorations, the solution can be placed on the lingual surface. In such cases, the improvement of color, as observed from the facial surface, will take longer to occur. Etching of the enamel is not indicated. Heat releases the oxygen more rapidly and is optional.[112] If heat is to be used to speed the chemical reaction, the source should be a dental curing light or equipment specially fabricated for bleaching procedures.

The patient must be asked if any tooth sensitivity is occurring. Development of discomfort is the single most important limitation to the number of applications per appointment. Appointments for subsequent treatment are scheduled 1 week apart to allow sensitivity to abate.[113] Because the number of appointments necessary to achieve the desired whiteness is variable,[114] the patient should be so informed and the fee per appointment set in advance. Generally, it is recommended that in-office

bleaching be followed by at-home bleaching to enhance the effect and to minimize rebound.[106,115,116]

Laser- or light-assisted in-office bleaching. Dental lasers and other intense light sources have been advocated by manufacturers for use in bleaching. It is claimed that they provide a powerful energy source to enhance the action of the hydrogen peroxide by promoting a more rapid release of the bleaching agent. Some concern has been raised over the safety of the use of lasers. Effects on hard tissues depend on the type of laser used as well as the exposure time. The temperatures created by high-energy light sources can be great and are influenced by the absorptive properties of enamel and dentin. The temperature level attained is also a product of the type of laser used.[117] The argon laser, used appropriately, generates very little temperature rise in the pulp,[118,119] whereas the diode laser can generate high temperatures that are detrimental to the pulpal tissue.[120]

One manufacturer recommends use of an argon laser with a wavelength of 488 nm for 30 seconds to accelerate the activity of the bleaching gel, although it does not seem to be effective.[121] The gel is left on the tooth for 3 minutes and then removed. This is repeated four to six times. Another technique uses a carbon dioxide (CO_2) laser after the procedure with the argon laser to encourage deeper penetration of peroxide into the tooth structure. Anecdotal reports have indicated that moderate to severe post-procedure pain and sensitivity may occur. Pulpal irritation or even necrosis has been demonstrated following CO_2 laser use.[122] The ADA does not recommend CO_2 laser use for bleaching.[123] Studies of laser bleaching to date have not indicated any better results than with other in-office techniques, and the results are possibly not as good as those obtained with at-home bleaching.[121]

It is important to remember that the hydrogen peroxide performs the bleaching, not the laser. The laser merely serves as a heat source to enhance the bleaching agent in the same way that a conventional curing light does. Its only potential advantage is its faster rate of supplying heat. Its disadvantages are the cost of the laser itself, which translates to higher patient fees, and the potential for damage to pulpal tissues and the surrounding periodontal tissues. A study evaluating the use of a light-emitting diode laser in combination with 35% hydrogen peroxide showed that the laser did not add any benefit in the final outcome; moreover, it increased tooth sensitivity.[113,124]

Other manufacturers have made claims about the use of high-intensity lights and their role in tooth whitening. The preponderance of evidence at this time does not

support their efficacy in producing a superior result.[125–132] One article indicated that the amount of ultraviolet radiation in a popular bleaching light exceeded the international standards for radiation safety.[132] Another study indicated that the use of a high-intensity light might lead to increased proinflammatory cytokine (IL-1β) levels.[133]

Dentist-supervised at-home technique

At-home bleaching requires a proper dental and medical history, clinical examination, radiographs of the teeth to be treated, and impressions for tray construction. After tray fabrication, an insertion appointment is required unless the tray can be fabricated during the first visit. Some dentists like to see the patient at weekly recall visits to assess progress and compliance, while others see the patient at the completion of treatment. As previously stated, initial treatment of only one arch has advantages of increasing compliance, decreasing sensitivity, and allowing a lower entry fee for the patient. Plans for treatment of sensitivity and determination of total treatment time should be communicated to the patient during the first visit.

At-home bleaching is the more commonly used bleaching process compared with in-office bleaching because it is easy to perform, it is usually less expensive for the patient, and there is strong evidence for its effectiveness.[134–141] It uses a custom-fit tray (Fig 16-23) with a 10% solution of carbamide peroxide (approximately equal to a 3.5% solution of hydrogen peroxide). Although the process requires longer contact time compared with the in-office bleaching technique,[142] it is safe, and the results are generally excellent. Products carrying the "ADA accepted" label have passed a rigorous set of safety and efficacy standards.[48,110–112,115,143–146] Manufacturers have offered carbamide peroxide in a variety of concentrations, ranging from 10% to 20% or higher, but the best combination of safety, limited side effects, and speed of action is obtained with a 10% solution. An initial survey indicated that 90% of the dentists surveyed used a 10% carbamide peroxide formulation for at-home bleaching of vital teeth.[147] However, the 15% formulation is also very popular.



Fig 16-23 The original tray design involved a nonscalloped, nonreservoir tray made from a cast of a good alginate impression.

More recently, hydrogen peroxide products have been introduced for day wear in trays. These products can be applied in a tray and range in concentration from 7.5% to 9.5% hydrogen peroxide. They are only active for 30 to 60 minutes (Table 16-2), so the wear time is shortened per day, but the number of days to achieve results compared with nighttime bleaching with carbamide peroxide is extended. Hence, instructions for hydrogen peroxide products typically recommend use for 1 hour or less per application.[148] Carbamide peroxide is active for up to 10 hours, with about 50% of the active agent being used up in the first 2 hours.[2,149] Carbamide peroxide is designed for nighttime application to achieve the maximum benefit,[20,150,151] but it can be used for 2 to 4 hours during the day. There have also been high concentrations of carbamide peroxide (35% to 38%) introduced for which the manufacturers' recommended tray application is only 3 to 15 minutes per day. A 35% carbamide peroxide product is approximately equivalent to a 9.5% hydrogen peroxide product. There is no research at this time demonstrating clear comparisons of this group of high-concentration carbamide peroxide products with traditional 10% carbamide peroxide tray bleaching.

Table 16-2 **Differences between carbamide peroxide (CP) and hydrogen peroxide (HP)**

	Carbamide peroxide	Hydrogen peroxide
Active time	2 to 10 hours due to carbopol amount	30 to 60 minutes
pH level	Neutral; elevates pH due to urea production	Low; pH stays constant during treatment
Lowest concentration	10% CP = 3.5% HP	6% HP = 17% CP
Effect of pH level	Caries process may not advance during bleaching due to elevation of pH to a basic level	pH does not change during treatment

Tray fabrication. Accurate impressions are critical to produce casts on which accurate vacuum-formed trays can be made. Because casts should be altered for tray fabrication, this often renders them unsuitable for other purposes. However, the

initial alginate impression, properly handled (impression kept wrapped in a wet paper towel, and the second pour accomplished within 45 minutes of the first pour), may be double-poured to provide an additional cast for other treatment needs.[152] Alternatively, polyvinyl siloxane impression materials provide multiple-pour accuracy over more extended time periods compared with alginate.[153] The cast for the bleaching tray should be trimmed to the thinnest and narrowest dimensions possible without damaging surfaces representing the teeth or periodontal structures (Fig 16-24). A model trimmer is used to trim the cast from the base rather than from the sides, until the vestibule is eliminated (see Fig 16-24a). The base of the cast should be as thin as 0.5 inch, or trimmed to a horseshoe shape, leaving only the maxillary or mandibular teeth and periodontal tissues remaining, with no palatal or tongue section included. The base should be flat, with the central incisors perpendicular to it (see Figs 16-24b and 16-24c). This makes it easier to adapt the vacuum-formed tray material around the teeth and avoids the development of folds and wrinkles during fabrication. If the palate or tongue section remains, it is helpful to drill a hole through that section so that the vacuum better adapts tray material in all areas of the cast.

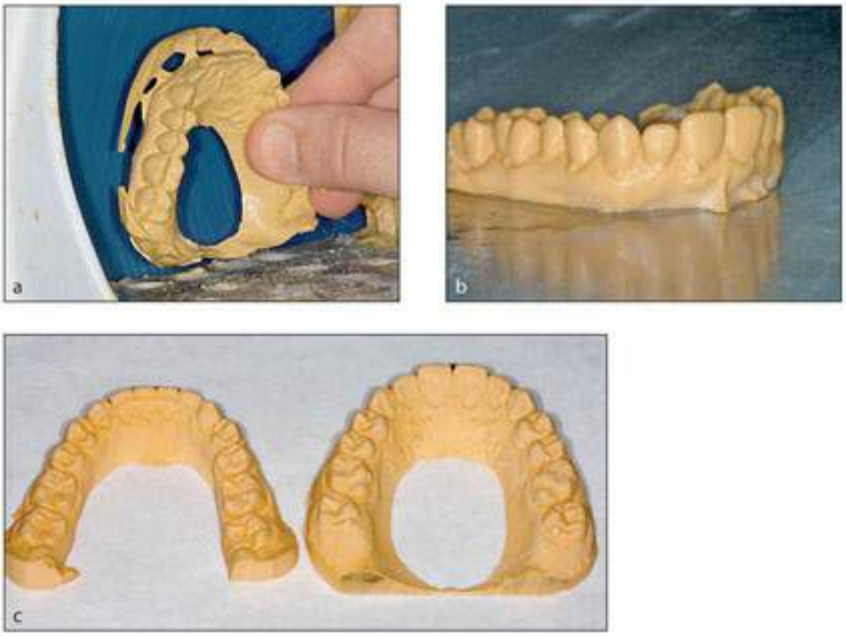


Fig 16-24 Trimming the stone cast for the bleaching tray is best accomplished using a nontraditional method. *(a)* The cast is trimmed from the base, keeping the central incisors parallel to the horizon. Gentle pressure is applied until the vestibule is removed. *(b)* In an ideally trimmed cast, there should be no vestibule, and the central incisors should be perpendicular to the tabletop to avoid wrinkles or folds in the tray material. *(c)* Casts work best in the vacuum former if there is no palate and no vestibule. In addition, a dry cast will allow the best adaptation of the tray material.

A number of materials have been used in tray fabrication, including materials used

in fabricating orthodontic position-ers, athletic mouthguards, provisional splints, and antisnoring devices. The original nightguard vital bleaching article[25,26] proposed use of a thick, semirigid material. The newer materials are thinner, softer, easier to shape and trim, and have reduced gingival and occlusal side effects. Additionally, one thermoplastic (boil and form) system uses a tray supplied by the manufacturer that is custom-fitted to the patient using heat. An alginate impression is not needed. This thermoplastic technique results in a tray formed in the patient's mouth at the first appointment.[79]

Decisions to be made about tray design include whether to scallop the tray borders and whether to add reservoirs. Some advocates of at-home bleaching were concerned that the agent would harm gingival tissues over time. They recommended a scalloped tray border, positioned 1 mm incisal/ occlusal to the soft tissue, to avoid contact of the tray or bleaching agent with the tissue[154,155] (Fig 16-25). This practice is not necessary for successful bleaching but may help when gingival tissues are very delicate.[156–158] Scalloping may be indicated for sticky, non–water-soluble materials, because they tend to adhere to the gingiva and may produce localized irritation. However, scalloping may be counterproductive if a water-soluble material is being applied, because the material is more rapidly washed out of this type of tray by saliva. Scalloping is not necessary when low concentrations of carbamide peroxide (10%) are used, but the higher the concentration of bleaching material, the more likely it is that there will be tissue irritation and the greater the need to scallop the tray so that it does not extend onto tissue. Hence, the tray design takes into consideration the concentration and type of material and the patient and his or her oral conditions and concerns. Additionally, scalloping can be annoying to the tongue or lips, depending on the patient and the teeth-to-tissue relationships. If tissue contact is preferred, the tray should not extend into undercuts or be tight enough to blanch tissue. The tray borders should be smooth. When tissue irritation during an at-home bleaching regimen has been reported, it often resulted from poor adaptation of a rigid tray and because the tongue, lips, and/or cheek have rubbed against an edge of the tray.[159]

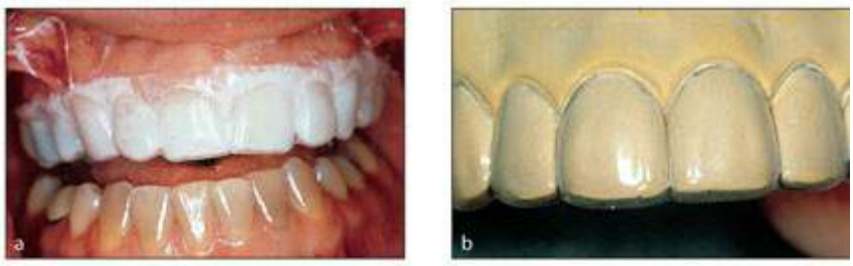


Fig 16-25 Depending on the patient, the bleaching agent, and the teeth, the tray can be designed

a s *(a) nonscalloped*, extending onto the tissue for low concentrations of peroxide, or *(b) scalloped* to avoid bleaching agent contact with the tissue. (White material was added to the tray for visibility in *a*.)

The patient should be able to close into the same occlusion with the thin tray in the mouth (Fig 16-26a) as without a tray (Fig 16-26b). If the anterior teeth are in contact in maximum intercuspation without the tray, then the anterior teeth should be in contact with the tray in place. The trays are too thin to adjust on the occlusal surfaces as in traditional appliances, so to achieve the proper anterior contact, the most posterior tooth molds of the tray should be removed (Fig 16-26c) and the occlusion reevaluated. The clinician should continue to remove the posterior tooth molds until anterior contact is achieved. Failure to maintain anterior contact will cause an increase in muscle activity[160] and may create headaches or muscle pain.



Fig 16-26*(a)* With the bleaching tray in place, the posterior teeth contact prematurely because of the arc of closure of the mandible, which keeps the anterior teeth from coming into contact. *(b)* Slight contact on the maxillary right central incisor in maximum intercuspation. *(c)* The mold of the terminal tooth is removed to allow further closure of the mandible. *(d)* Enough teeth are removed to allow the tray to contact the anterior teeth in the same manner as the natural teeth.

Foam liners for the tray have been advocated to hold the bleaching agent evenly against the teeth by preventing flow of the bleaching agent to the incisal portion of the maxillary tray or the gingival portion of the mandibular tray. Despite manufacturers' claims that the inclusion of foam liners gives a more rapid bleaching effect, research has failed to demonstrate an advantage.[161] Some techniques incorporate a reservoir on the facial intaglio surface of the tray to hold a greater volume of the agent in the target areas of the teeth to enhance the process[112,162] or to allow bleaching gel to flow from areas of high concentration to areas of reduced

concentration.[163] The theory is reasonable, but there is little evidence that the creation of reservoirs actually improves bleaching efficacy.[137] It is assumed that the solution degrades at the same rate whether it is present in greater volume in a reservoir or not. Some clinicians do not advocate placement of reservoirs in the tray, because this seems to waste bleaching material, increase the difficulty of tray fabrication, and reduce the comfort of the tray.[27,52,164–166]

Reservoir spaces are formed by using a manufacturer-supplied, light-polymerized resin, placed 0.5 to 1.0 mm thick on the facial surfaces of the teeth of the cast. The resin spacer should terminate 1.0 mm short of the gingival area and should not extend into the embrasures. Nail polish, die spacers, or tinfoil may also be used to form reservoir spaces. The reservoir design should allow tray borders to contact the tooth. Spacers should not be placed in areas of occlusion (such as incisal aspects of mandibular anterior teeth or lingual aspects of maxillary anterior teeth), because the contact of the opposing teeth will displace the material from the tray.

There may be some benefit to having the extra space provided by the reservoirs, such as to help in seating the tray and to avoid pressure on the teeth, which may result in less sensitivity, especially if a very viscous solution is used.[167] One study, however, has indicated no difference in patient comfort with or without tray reservoirs,[166] and several studies have shown no clinical difference in efficacy between reservoir and nonres-ervoir trays.[137,165,168]

When one tooth is darker and the other teeth are reasonably acceptable (see Figs 16-15a to 16-15c), a single-tooth bleaching tray can be fabricated in the same manner as described for the nonvital tooth (see previous section). This tray design works well with vital teeth also.

Patient instructions for at-home bleaching. Patient instructions should address expectations and the course of treatment, the technique for applying the bleaching agent, the frequency and length of time for wearing the tray, tooth sensitivity/tissue irritation problems, interim appointments, and variations in the total fee related to the course of treatment. It is critical that the patient understands the process and can make appropriate adjustments in the protocol (eg, discontinuing tray wear for 1 or 2 days if sensitivity begins to develop). The patient must recognize and inform the dentist of any developing problems early in the process. While this is a supervised procedure, most of the process occurs away from the dentist, so the patient becomes the progress monitor. The patient should be instructed to place enough bleaching agent into the tray to cover the facial surfaces of the target teeth. This includes the most posterior tooth visible when the patient smiles, laughs, or talks. Seating and

removing the tray should be demonstrated to ensure that the patient is able to do so without undue difficulty or harm to the tray or the oral structures. To avoid tissue injury, the tray should be "peeled" from the second molar area rather than being "dug out with fingernails" in the canine region (Fig 16-27).



Fig 16-27 The tray is removed by grasping the most terminal tooth rather than digging fingernails around the edges of the tray and into soft tissue.

Instructions for tray wear. The correct time for wearing of the tray depends both on the product being used and the patient's lifestyle. Ideally, carbamide peroxide is best worn overnight for maximum benefit of the product and compliance by the patient. Hydrogen peroxide is indicated for daytime wear because of the short active period of 30 to 60 minutes. While carbamide peroxide is ideally used overnight, the patient also can use it during the day, but the patient should understand that he or she is to wear the tray daily for 2 to 4 or more hours. Because the material maintains its potency for several hours, it is wasteful to wear it for fewer than 2 to 4 hours. One study demonstrated that 50% of the active agent remained after 2 hours.[2,149,168] The best option is probably for the patient to wear the tray during sleeping hours. Although the agent has lost most of its potency after 5 hours, compliance is much better when the bleaching treatment becomes part of a regular nightly routine. After the loaded tray is seated, the patient may use a damp cloth or finger to wipe the areas adjacent to the tray borders to remove any excess bleaching material. However, swallowing low concentrations of carbamide peroxide is not harmful as its original use was as an oral antiseptic to aid in wound healing. The patient should be reminded to rinse and gently brush the tray after each session before storing it in a cool or room-temperature environment until the next bleaching session.

The amount of time required for the bleaching process varies (Table 16-3). While some readily discernible improvement may occur within 2 to 14 days, it may take as long as 6 to 12 months.[159] The time required depends on the type of discoloration, patient compliance, and whether or not any tooth sensitivity occurs. It may be helpful for improving compliance to recall the patient after 5 to 7 days to check progress. The patient can be shown the shade tab representing the original color to contrast it with the new whiteness of the teeth, or the patient can compare the treated maxillary

arch with the untreated mandibular arch. At an interim appointment, tray adjustments may be necessary, and digital photographs can be taken to compare with the prebleaching photographs. The protocol can be reviewed to ensure that the patient is following it correctly. Alterations in protocol can be made to eliminate or minimize any problems or to speed the process. Some teeth may respond to bleaching more rapidly than others. More severely discolored and less responsive teeth may need more sessions of bleaching than others.[48,169] Some teeth with white spots exhibit a splotchy look as different portions of the tooth respond at different rates. The patient should be encouraged to continue treatment until the remainder of the tooth achieves the same color. The splotchy areas tend to abate on completion of treatment. Considerable whitening of the teeth is routinely predictable (Figs 16-28 to 16-33).

Table 16-3	**Treatment times for tray bleaching**
Normal teeth	3 days to 6 weeks (average of 2 weeks)
Nicotine-stained teeth	1 to 3 months
Tetracycline-stained teeth	1 to 12 months (average of 3 to 4 months)



Fig 16-28 Yellow-brown stains usually respond quickly to at-home bleaching. *(a)* Prior to bleaching. *(b)* After 1 month of bleaching the maxillary teeth using 10% carbamide peroxide in a tray.

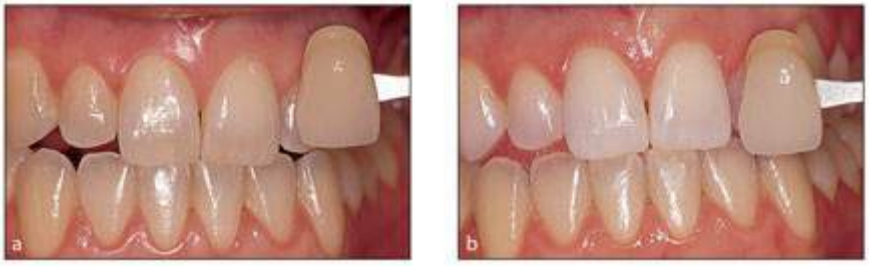


Fig 16-29 Many gray stains respond well to at-home bleaching. *(a)* Prior to bleaching. *(b)* Results after more than 4 weeks of bleaching the maxillary teeth with 10% carbamide peroxide.



Fig 16-30 White spots become less noticeable when the teeth are whitened. *(a)* Prior to bleaching. *(b)* After bleaching the maxillary teeth for about 6 weeks with 10% carbamide peroxide. Note that the white spot is unchanged but less noticeable.



Fig 16-31 Yellow-brown stains can be removed easily with at-home bleaching with 10% carbamide peroxide. *(a)* Prior to bleaching. *(b)* After bleaching for 6 weeks.



Fig 16-32 Yellow stains can be removed more readily from the coronal portion than from the root of the tooth. *(a)* Prior to bleaching. *(b)* After tray bleaching the maxillary teeth with 10% carbamide peroxide for 6 weeks. The roots retain their yellowish coloration, likely because of differences in composition and dentin tubule size and density between root dentin and coronal dentin.[54]



Fig 16-33*(a)* Unknown brown discoloration, possibly from trauma to a primary tooth, prior to bleaching. *(b)* After tray bleaching for about 6 weeks with 10% carbamide peroxide.

Consumer-managed at-home technique

Over-the-counter (OTC) products are available in the form of tray systems, trayless systems, and paint-on products. The fundamental problem with OTC products is that there is no examination and diagnosis of the cause of discoloration. Hence, any effective bleaching treatment may mask symptoms of another problem. Discoloration could indicate that the teeth are non-vital or could be caused by dental caries, internal resorption, or dark, stained, or discolored restorations. It is advisable for patients considering bleaching to have a proper clinical and radiographic examination performed by a dentist, even if they are considering using OTC products (Box 16-9).

Box 16-9	**Safety concerns for OTC bleaching**
• Lack of diagnosis for proper treatment (pathology masked; bleaching not indicated) • Potential for less esthetic outcome (restorations not identified; endodontic status not known) • Unknown safety of higher concentrations (no research above 15% CP or 6% HP; issues with unsupervised use) • Unknown quality of some products (pH, allergenic ingredients, etchants, other ingredients) • Patients may not receive any or maximum benefit avail able for whitening (due to shortened treatment time for difficult stains and discolorations, ineffective products)	

CP–carbamide peroxide; HP–hydrogen peroxide.

In the past, OTC bleaching agents did not prove very effective. Although the initial cost of the agent or system was distinctly lower than dentist-prescribed methods, its ineffectiveness led to repeated purchases of the product, so the final cost was significant and the results often unsatisfactory. Additionally, early reports cited detrimental effects of tooth structure erosion from overuse of poorly manufactured or formulated solutions.[170]

Recently introduced products have proven to be much more effective in whitening teeth. While the specific mode of application varies among the products, the manufacturers have made the products relatively easy for the patient to apply. One whitening system utilizes adhesive strips containing a relatively high concentration (approximately 6% to 14%) of hydrogen peroxide. The adhesive strips are designed to hold the peroxide against the teeth to allow it to effectively bleach. Patients are instructed to wear the strips for 30 minutes twice a day for maximum benefit. Studies

have shown the whitening strips to be effective[171,172] without significant tooth sensitivity or tissue irritation. They are relatively convenient and less expensive than dentist-supervised at-home techniques. The strips may be difficult for some patients to adapt to the target teeth, especially if the teeth are malaligned. In addition, the strips do not extend past the canines in many people's mouths, so posterior teeth are not whitened and the final result may not be pleasing in patients with a broad smile. Few studies have compared strips with daytime tray bleaching and nighttime tray bleaching using the same color measurement criteria as in previous tray-bleaching research. However, one study did show that daytime wear of hydrogen peroxide in the tray was equivalent to that of strips, but nighttime wear of carbamide peroxide was more efficacious.[138] There are many other OTC products on the market that change frequently, from paint-on products to different versions of strips. The original concentrations of many researched OTC products have been altered, so it is very difficult to find clinically useful research.

There are a few studies that have compared OTC strips using hydrogen peroxide with tray bleaching using carbamide peroxide and in-office bleaching using hydrogen peroxide. To obtain a six-shade color change on a VITA shade guide required 7 nights of tray bleaching with 10% carbamide peroxide, 16 days of a 6% hydrogen peroxide strip (OTC) worn twice a day, or 3 in-office treatments of 38% hydrogen peroxide.[138] If the three bleaching techniques are used for 2 weeks, then tray bleaching is most efficacious, similar to in-office but significantly better than OTC strips.[173]

Myriad other OTC products, from toothpastes to chewing gums, claim tooth whitening with little or no research to support their claims.[174] A great many of these products produce whitening by removing extrinsic or surface stains (with abrasives or peroxides), preventing staining (with chemicals), or merely physically removing food debris from already-white teeth (chewing gums).

Manufacturers continue to develop new OTC products. Although new chemistry may be introduced, their focus has been primarily on delivery systems that are easier and faster for the patient to apply. However, the most critical challenge with the use of OTC products is the lack of a proper prebleaching examination to determine if bleaching is appropriate for the patient. The lack of a proper examination may cause pathology to be masked by bleaching and appropriate treatment delayed.

A general summary of the three types of bleaching is that overnight tray bleaching produces the best results,[69] followed by daytime-wear tray bleaching. OTC bleaching strips are equal to in-office and similar to daytime-wear tray bleaching.

In-office bleaching should be used with tray bleaching to complete the process.

Techniques for tetracycline-stained teeth

Patients whose teeth have been stained by tetracycline ingestion present a very difficult esthetic challenge. Tetracycline can be deposited in fetal tooth buds if ingested by an expectant mother in the third trimester of pregnancy or by a child during the tooth-formation years, between the ages of 3 to 4 months and 7 to 8 years.[175] Tetracycline may also be deposited in the teeth in early adults if it is taken on a long-term basis for skin conditions, such as acne, especially during the formation of secondary dentin, during growth periods, and after trauma.[37,38]

Tetracycline has several different analogs (eg, tetracycline, doxycycline, oxytetracycline, minocycline, chlortetracycline, demeclocycline) that may produce various intensities of gray, blue, brown, and yellow in the teeth. If seemingly normal yellow teeth are not responsive to the conventional 2 weeks of bleaching treatment, they may actually be tetracycline stained.

Recent research has shown that tetracycline-stained teeth may respond to bleaching treatments but at a rate different from that of teeth stained by other agents[176,177] (Figs 16-34 and 16-35). Whereas the normal bleaching time is 2 to 6 weeks, some tetracycline-stained teeth may require 2 to 12 months of daily treatment to achieve a significant improvement.[30,178] Tetracycline-stained teeth do not generally lose all discoloration. Although the tooth color is often much improved, teeth may retain some grayish-blue tint.



Fig 16-34 Tetracycline stains are tenacious; results come more slowly than with other types of stains. *(a)* Prior to bleaching. *(b)* Treatment midpoint. *(c)* After approximately 7 months of at-home bleaching.



Fig 16-35 A generalized tetracycline stain. *(a)* Prior to bleaching. *(b)* After several weeks. *(c)* After more than 8 months of at-home bleaching with 10% carbamide peroxide.

Research on the longevity of color change achieved after bleaching tetracycline-stained teeth indicates that most patients will have some degree of lightening and that 8 in 10 patients can expect to retain that lightening for at least 1 year. Even those patients who experience some regression indicate that they were glad they bleached their teeth and would do it again. A follow-up study showed a very high patient satisfaction rate at 7.5 years after completion of bleaching.[30]

There are several factors to consider when bleaching tetracycline-stained teeth for extended periods.[110] First, the location of the stained area has a great influence on the prognosis for success. A tooth generally lightens from the incisal to the gingival area because the tooth gets progressively thicker from incisal to gingival. Teeth heavily stained in the gingival third have the poorest prognosis for complete lightening. The further toward the incisal edge the stain resides, the better the prognosis. In any situation, absolute predictions of success are unrealistic. Patients must understand that each discoloration responds differently and that they may not see significant results in the first few months.

A practical consideration is the amount of bleaching material necessary for extended treatment and the appropriate fee for service. Practitioners may choose either to increase the total fee or use the initial fee for normal bleaching treatment with a monthly fee for each additional month of treatment. The monthly fee is dependent on the amount of material the patient uses monthly (as determined by recording dosage in a log form) and the frequency of office visits (usually 1 to 2 months apart). Patients can then pay as they go for extended treatment if there is continued, albeit slow, improvement. Another more affordable option that has been shown to be effective is to bleach tetracycline-stained teeth using 6.5% hydrogen peroxide strips.[179]

Generally, teeth severely stained in gingival areas are candidates for porcelain veneers or crowns if maximum ideal esthetics is desired, but it is generally best to attempt bleaching first. Bleaching may achieve adequate results and avoid the need for veneers, even if the result is not quite as esthetic. The bleaching may have only a limited lightening effect, but it can reduce the amount of opacity necessary in the veneer for masking the stained tooth structure. This is especially important if the teeth are misaligned, because the veneer preparation may be extended closer to or into the darker dentin, making it difficult to obtain an even translucency in the veneers when areas of color contrast exist in the tooth. Even if the bleaching

treatment is ineffective, the patient is aware that the most conservative avenues have been attempted first and is assured that porcelain veneers or porcelain-fused-to-metal or all-ceramic crowns are the best remaining options. If the teeth under the veneers discolor, or if the veneers have been placed on non-bleached teeth, the natural teeth can be bleached under the veneers from the lingual.[180] This technique can easily lighten previously bleached teeth back to their original shade and hence return the veneers to their original appearance. Depending on the translucency of the veneers placed on nonbleached teeth, there can be some moderate lightening appearance in the veneers if they are relatively translucent.

Dentists and physicians are well aware of the effect of tetracycline ingestion on dental esthetics,[25,34,35] although the results of tetracycline absorption in teenagers treated for acne has only recently been reported.[181–189] Tetracycline is still the drug of choice for outpatient treatment of Rocky Mountain spotted fever and is the most widely prescribed drug for acne. Dentists should consider bleaching a reasonable treatment option for the discoloration resulting from this treatment if the patient is willing to comply with extended treatment and does not expect total elimination of the stain.

Factors affecting both the in-office and at-home bleaching processes

Several factors must be considered carefully before bleaching is initiated and then controlled during the process to ensure maximum benefit.

Surface cleanliness

All surface debris must be removed to distinguish intrinsic from extrinsic staining and to ensure that the agent has maximum contact with the tooth surface. However, bleaching should be delayed for several days after dental prophylaxis to allow any gingival or tooth sensitivity related to the prophylaxis to abate. Some stains begin externally but absorb internally. Nicotine is an example and is a very tenacious stain to remove both externally and internally (Fig 16-36). However, bleaching for a period of 1 to 3 months with 10% carbamide peroxide can remove the internal portion while the prophylaxis can remove the external stain.



Fig 16-36 Nicotine stains. *(a)* Prior to bleaching. *(b)* After bleaching.

Concentration of peroxide

The higher the concentration of peroxide, the more rapid the lightening effect up to a certain point, although the effect is not linear (ie, 20% is not twice as fast as 10%).[43,190] The limiting factor is how fast the tooth can change color. Once the maximum change rate of the tooth has been achieved, the additional concentration contributes only to sensitivity.[191] Hence, conversely, the higher the concentration, the greater the chance for sensitivity. In-office bleaching materials are usually supplied in concentrations of 35% hydrogen peroxide, although concentrations range from 15% to 50%.

Temperature (in-office bleaching)

The higher the temperature of the bleaching solution, the faster the rate of oxygen release and, therefore, the faster the rate of the chemical reaction. An increase of 10°C doubles the rate of chemical reaction.[192,193] However, this does not necessarily alter the rate of the tooth color change.[194] The rate of tooth color change is more dependent on the tooth than the material used for bleaching. Teeth change color at different rates and progress to a different whiteness. Additionally, temperatures elevated to an uncomfortable level may result in tooth sensitivity or even irreversible pulpal inflammation.[195] Bleaching materials are always applied without anesthesia to avoid overheating the tooth. Nonvital teeth should not be heated to a temperature higher than that acceptable for a vital tooth.

pH

The pH of the bleaching material can influence both the effects on the teeth from bleaching as well as the concern for existing decay. Enamel caries starts when the pH is below 5.5; dentin caries is initiated when the pH is below 6.8. Material Safety Data Sheets (MSDS) list hydrogen peroxide products with a pH around 5. Most carbamide peroxide materials approved by the ADA have a pH of approximately

7.[196] Materials with a significantly lower pH can cause tooth surface alterations by their acidic nature. However, carbamide peroxide breaks down quickly into hydrogen peroxide and urea when applied. Urea is primarily responsible for raising the pH in the oral cavity above 8 for a number of hours. Therefore, there is a potential for retarding caries activity during bleaching with carbamide peroxide.

Time

In addition to the concentration, the degree of bleaching is directly related to the amount of time the bleaching agent is in contact with the tooth. The longer the contact, the more lightening will occur (until the plateau of the tooth color change is reached); however, the longer the bleaching agent is in initial contact with the teeth, the greater the likelihood of tooth sensitivity.[197] Interestingly, the greatest sensitivity seems to occur in the first 2 weeks of treatment. After this initial time period, sensitivity does not continue to increase on all patients but tends to be more episodic.

Additives

Many of the peroxides have additives to alter their handling characteristics or patient acceptability of the product. Materials may be added to liquid hydrogen peroxide to form a gel for easier handling and safety, but this may also alter the efficacy of the material. Carbamide peroxide may also have various ingredients added to promote thickness, stickiness, or viscosity. Carbamide peroxide can have many different base vehicles, including variations of glycerin, glycol, and toothpaste base materials. Additionally, it is manufactured in many flavors, and some of the flavoring additives can cause mild adverse responses in certain patients.[198] Some products may contain gluten, making them inappropriate for celiac disease patients. Some flavors like banana may have cross-linkage to latex allergies.[199] Some mint flavors may promote aphthous ulcers or cause gingival burning.[200,201] The most significant additive to carbamide peroxide is carbopol. This ingredient thickens the material and prolongs the release of peroxide over time.

Safety factors

Safety concerns include the potential for tooth/pulpal problems, irritation of periodontal or other oral tissues, and systemic effects. The primary safety issue is the concern for masking existing pathology because of the potential of bleaching the only symptom, which is a discolored tooth or teeth. It is important for the patient to

have a proper examination including radiographs prior to the initiation of any type of bleaching.

Tooth Sensitivity and Pulpal Concerns

Short-term pulpal response varies from patient to patient and even from tooth to tooth. Peroxide penetrates through the tooth to the pulp in a matter of 5 to 15 minutes.[72,202] The peroxide molecule is of such small size that it easily passes through the interstitial spaces of the tooth, rather than traveling down the dentinal tubules. Therefore, bleaching can produce sensitivity.[31] However, the pulp remains healthy and the sensitivity is completely reversible.[192,202,203] Patients undergoing at-home bleaching must also be informed that minor sensitivity may occur in as many as two-thirds of patients. Double-blind studies have reported sensitivity in 25% to 70% of patients. However, these studies also show that 20% to 30% of patients report sensitivity when a placebo is used, and 18% report sensitivity just from wearing the bleaching tray without any gel.[33] Manufacturers have included potassium nitrate and fluoride or amorphous calcium phosphate (ACP) in the bleaching gel. These new bleaching formulas have been shown to have varying degrees of effectiveness in decreasing tooth sensitivity.[139,204,205]

If sensitivity occurs, there are a number of approaches available involving either passive treatment or active treatment to resolve the sensitivity (Box 16-10). Passive treatment involves shortening the duration or frequency of treatment or interrupting the process for a day or more to allow the teeth to recover. The procedure can then be resumed.[206] Active treatment involves the application of medicaments using the same bleaching tray. Historically, fluoride has been applied for sensitivity. Fluoride acts as a tubule blocker to limit the fluid flow to the pulp.[207] It has been suggested that fluoride treatment prior to initiating bleaching may reduce sensitivity; however, there is little research showing this to be the case. Bleaching sensitivity is not directly related to tubule blockage, which is the mechanism of action of fluoride.

Box 16-10	**Treatment of patients with maximum sensitivity**
• Do not initiate bleaching immediately after the prophylaxis. • Brush with a potassium nitrate–containing toothpaste for 2 weeks prior to initiating bleaching . • Wear the empty tray 1 or 2 nights to become familiar with the feel and pressure of the tray.	

- Wear tray with potassium nitrate toothpaste (without sodium lauryl sulfate) 1 or 2 nights.
- Begin bleaching initially with shorter daytime periods using 10% carbamide peroxide with potassium nitrate and fluoride in the product.
- Continue brushing with desensitizing toothpaste during bleaching.
- Use tray delivery of potassium nitrate desensitizing materials 10 to 30 minutes in the tray as needed during treatment. Use a professional product or OTC desensitiz ing toothpaste with 5% potassium nitrate.
- Alternate or skip bleaching days, if sensitive.
- Avoid acidic drinks, cola, and fruit juices, which remove the smear layer.

A more direct treatment is the application of 3% to 5% potassium nitrate gel in the tray. Potassium nitrate preparations are available from several bleaching agent manufacturers.[208] Potassium nitrate penetrates the tooth to the pulp much like the peroxide molecule and has an anesthetizing or sedative effect on nerve transmission.[209,210] Potassium nitrate is found in many desensitizing toothpastes but generally takes 2 weeks to be effective via tooth brushing. However, application of a potassium nitrate toothpaste in the tray for 10 to 30 minutes before or after bleaching can reduce or eliminate sensitivity in many patients.[211] The best toothpaste to use is one that contains 5% potassium nitrate but does not contain sodium lauryl sulfate (SLS), which is a detergent foaming ingredient. SLS has been associated with removal of the smear layer and increased occurrence of aphthous ulcers.[201]

Often, treatment for sensitivity is a combination approach involving alteration of duration or frequency of treatment and use of medications, including anti-inflammatory medications, desensitizing toothpastes, and desensitizing medicaments applied in the tray. A combination of potassium nitrate and fluoride may be used in the tray to optimize the desensitizing results.[208,212,213] Questions have been raised about the effect of bleaching on the structure of the tooth itself.[214] Recent studies have shown that low-pH solutions can produce a detectable loss of calcium from the surface enamel, along with a slight loss in surface hardness to a depth of approximately 25 μm.[27,215–217] However, this loss has not been shown to be significant because the surface quickly remineralizes after the procedure is completed.[218–223] Moreover, a study concluded that "no obvious morphological or chemical composition alterations of enamel surface were detected in the neutral or alkaline bleaching solutions."[224] In fact, there is less change in the calcium content of

the tooth and surface hardness from 6 hours of bleaching with 10% carbamide peroxide than when a carbonated drink is consumed in a 2- to 3-minute period. No noticeable change in the surface luster and topography is seen clinically.[225]

Soft Tissue Response

The more powerful in-office bleaching agents (30% to 35% hydrogen peroxide) can easily produce tissue burns, turning the tissue white.[226–228] If the exposure is limited in duration and quantity, it is quickly reversible with no long-term consequences. Rehydration and application of an antiseptic ointment (eg, Orabase B, Colgate Oral Pharmaceutical), baking soda, calcium hydroxide powder mixed with distilled water, or zinc oxide–eugenol provisional cement (eg, TempBond) quickly returns the color to the tissue and removes the burning sensation (see Fig 16-22), reassuring the patient that the problem is not permanent.[66] Nevertheless, it can cause significant temporary discomfort and some alarm when first seen. It is important to protect soft tissues with rubber dam or other means to avoid tissue burns.

Although soft tissue irritation during at-home bleaching has been reported,[163] the irritation is most likely the result of an ill-fitting tray rather than the agent itself.[229] Reports of harmful effects to soft tissues from hydrogen peroxide indicate that the effects resulted from dosages and exposure times that greatly exceeded those prescribed in any at-home bleaching technique. At-home agents containing 10% carbamide peroxide are not potent enough to produce significant or long-lasting effects on the soft tissues.[230] However, the higher the concentration of carbamide or hydrogen peroxide used in the tray, the greater the need to avoid soft tissue contact by scalloping the tray.

Studies have indicated that approximately one-third of patients experience no detectable side effects after bleaching, and the other two-thirds experience only minor transitory tooth sensitivity and/or tissue irritation of short duration.[32] When examined at the cellular level, these effects on the soft tissue are less than or equal to those produced by commonly accepted dental medicaments, such as eugenol and end-odontic sealers.[230] The toxicity and mutagenicity of hydrogen peroxide are dose related. Concentrations used in the at-home bleaching technique are not sufficient to be of concern. A low dose of hydrogen or carbamide peroxide over a long period actually allows the cells of the oral tissues to adjust to the dosage even if it is increased beyond the original tolerable dosage. In the long history of these materials

involving tissue contact in patients ranging in age from infancy[231,232] to old age, there has been no demonstrated health problems of any type.[233,234]

Systemic Effects and Responses

There is more concern about possible adverse effects of at-home bleaching agents compared with the in-office agents, although the at-home concentrations are far lower than those of the in-office bleaching agents. In-office bleaching agents are carefully controlled and placed on the teeth only, avoiding contact with soft tissues. The patient swallows no solution. Very little, if any, of the agent is absorbed systemically. At-home bleaching agents, applied in the tray, unavoidably contact soft tissues in many areas over several hours each day. Additionally, it is likely that the patient will swallow small amounts of very dilute hydrogen peroxide during the bleaching procedure.[146] This has not proven to be a health concern or systemic factor, however. Some patients may have a throat irritation or may have tissue sensitivity to different flavors. Although very high concentrations of some forms of peroxide are mutagenic,[235,236] physiologic mechanisms quickly repair any limited damage that might occur.[237] Low levels of hydrogen peroxide do not cause significant problems.[238] Carbamide peroxide is used for the treatment of candidiasis in newborn infants.[231] In fact, hydrogen peroxide has been approved as safe for use as a human food additive. The Food and Drug Administration recognizes 3% hydrogen peroxide and 10% to 15% carbamide peroxide as GRAS (generally recognized as safe) for oral use, but there is very little research on systemic effects related to the use of carbamide peroxide concentrations higher than 10%. Another paper notes that the human body creates peroxide in amounts far greater than that typically used in tray bleaching, and because we naturally create peroxide, our bodies have salivary peroxidases that remove the peroxide.[239]

The conclusion, after decades of use and extensive research, is that the use of low concentrations of hydrogen peroxide for bleaching teeth is safe.[222,240–243] A recent examination of all the research on bleaching by the European Scientific Commission on consumer products suggested the following:

- Bleaching with low concentrations of peroxide is safe.
- OTC products are not recommended, but if the patient opts to use these products, he or she should have a proper oral examination by a dentist, then obtain a prescription to purchase OTC-type products.

- The maximum recommended concentration for products is 6% hydrogen peroxide (which is approximately equivalent to 17% carbamide peroxide).[244,245]

Other Considerations

Effect on restorations

Bleaching has little or no effect on most of the common restorative materials.[142,215,216] There are no reports in the literature indicating that bleaching exerts any negative impact on existing restorations that would require their replacement.[246] Perhaps its most significant effect is that it lightens teeth enough that previously placed restorations may appear comparatively dark, leading to the consideration of replacing restorations for esthetic rather than functional reasons. Bleaching has no effect on porcelain. It does encourage the release of mercury from some types of amalgam restorations.[247,248] The clinical significance of this is not known.[249] The surface of some types of resin composite is roughened slightly, and the hardness may be slightly decreased, but neither effect is clinically significant.[250–254] Moreover, the higher the concentration, the more evident the effects.[255] Bleaching does affect methyl methacrylate provisional restorative materials, causing them to yellow slightly.[256] Although bleaching releases much oxygen into the tooth, the bond of existing restorations is not weakened. There are no contraindications for bleaching in the presence of existing bonded restorations. However, as previously stated, the oxygen-rich tooth structure does not provide a good surface for bonding, because the released oxygen hinders the polymerization of the resin.[257] There may also be changes in the surface morphology that affect the bonding.[218,258–260] A delay of 1 to 2 weeks or more following the bleaching process allows this effect to dissipate so that bonding can effectively be performed.[63] This also allows the tooth shade to stabilize before selection of the restorative material shade.[69] A drying agent, such as acetone, can be used to diminish the oxygen in the outer layers of the tooth if there is concern that oxygen has been retained in the surface.[66]

Alternatives to bleaching

The alternatives to bleaching are more aggressive in relation to the tooth structure and/or to the gingival tissues. These procedures include microabrasion,

macroabrasion, bonded resin composite veneering, porcelain veneers, and porcelain or metal-ceramic crowns. Although all of these are legitimate treatment techniques with proven efficacy, they rely on the removal of tooth structure and/or the addition of materials that have finite replacement life cycles.

Microabrasion and macroabrasion

One decision in the esthetic treatment planning for a patient is whether to remove discoloration by bleaching or by removal of tooth structure. For example, fluorosis is often amenable to treatment with microabrasion. *Microabrasion* is a process in which the tooth surface is subjected to a combination of an acid and an abrasive[261–263] (Fig 16-37). The acid removes mineral content, leaving the outer 22 to 27 μm weakened enough so that the abrasive, with which the acid is mixed, quickly removes the stained outer surface of the tooth. Although the amount of enamel removed is very limited, this abrasive technique alters the outer surface of the enamel as the undesirable coloration is removed or at least modified. The abrasive, usually suspended in a water-soluble gel containing a low concentration of hydrochloric acid, is applied to the enamel with a rubber cup or stiff bristle brush for 20 to 30 seconds.[262] Superficial discolorations usually can be removed from the enamel surface, but it is not possible to know in advance whether the discoloration is superficial. Should the discoloration extend to the dentin or become more visible, a resin composite restoration will need to be placed to seal the defect and restore contour. The choice of resin composite poses a shade-selection dilemma as to whether the material should match the original tooth color or the color the tooth will be. It is best to bleach first unless the tooth enamel is of a chalky texture. Then when abrasive techniques are initiated, the tooth shade is already established should a resin composite restoration be required. Bleaching has been estimated to remove 80% of brown discolorations.[25] White discolorations are not removed but may be much less noticeable when the surrounding area of the tooth is lightened. If bleaching is unsuccessful, then the more aggressive techniques can be initiated. The patient should be informed of the treatment options (bleaching, abrasion, bonding composite or porcelain) before treatment and should understand the different fees for these procedures. A combination of microabrasion or macroabrasion and bleaching may be sufficient to bring about the desired result.



Fig 16-37 Differences between microabrasion and macroabrasion. *(a)* The patient presents with superficial white and brown stains. The discoloration will be bleached. *(b) Microabrasion* involves the application of hydrocholoric acid in a prophylaxis paste to first soften the enamel and then use of a geared-down slow-speed handpiece with heavy pressure and a special applicator to abrade the softened enamel. *(c) Macroabrasion* involves the use of fine diamond and carbide burs in a high-speed handpiece with water spray and light pressure. This removal system is similar to polishing a composite restoration. *(d)* The outcome is very similar. Microabrasion was completed on the maxillary right cental incisor and macroabrasion was completed on the maxillary left central incisor. There are some subtle surface differences. Caution should be used to avoid moving the line angles of the tooth proximally, which would give the tooth a wider appearance. (Courtesy of Rod Mackert, Augusta, Georgia)

Another form of abrasion is *macroabrasion*. In this technique, the abrasion can be achieved by applying a fine diamond or carbide finishing bur to the enamel at relatively low speeds and very light pressure (see Fig 16-37c). Some of the same instruments that are used to contour and polish composite restorations may also be used on enamel.[264] The effect is much the same as with the acid/abrasive technique. When using a high-speed handpiece, water should be applied to maintain the coloration of the teeth. As some teeth dehydrate, more subsurface defects may become visible; for these, treatment is not indicated. Air-abrasive techniques can also be used quite effectively for microabrasion or macroabrasion. Only fine abrasive particles should be used to avoid overreduction of the tooth. Overreduction can be corrected by adding resin composite to the undercontoured areas. With any technique used, it is important to maintain the correct position of the line angles of the teeth. If the mesial and distal line angles are moved further apart, then the tooth will appear wider. Facial embrasure spaces should be maintained to keep the individuality of the teeth.

All three techniques (diamond-bur surface alteration, chemical/physical microabrasion, and kinetic-energy preparation [air abrasion]) rely on the selective reduction of the outer surface of the tooth. While they may be very effective in removing a relatively shallow discolored area, they are not intended to eliminate overall discoloration, especially if it involves the underlying dentin. However, they are valuable techniques for treating limited areas and for supplementing tooth bleaching.[264] For patients with a limited demand for esthetic improvement in the color of the teeth, abrasion alone may satisfy the patient's needs. If abrasion techniques do not produce completely satisfactory results, more extensive and invasive procedures may be performed.

Bleaching before and after placement of veneers

Discolored teeth may be treated with bonded restorations to cover or disguise the underlying stain, and it may be possible to place the material with little or no tooth reduction. The esthetic restorative material, such as resin composite or porcelain (see chapter 17), is bonded to the tooth to provide good color and form. However, the natural appearance of the veneered surface depends strongly on some light reflecting from the underlying tooth structure. If the tooth structure is too darkly stained, the discoloration will show through the translucent resin composite or ceramic veneer. The only solution is to make the veneering material thicker and less translucent. To avoid overcontouring the tooth/restoration, it is necessary to remove more of the facial and proximal tooth structure to achieve the esthetic and functional objectives. Successful bleaching therapy can minimize the need for a thicker veneer and more tooth reduction.

Bleaching is possible even for teeth that have already received veneers.[180,226] Application of a bleaching agent to the lingual surface can alter the apparent color of the translucent restoration on the facial surface (Fig 16-38), thereby improving esthetics.



Fig 16-38 Bleaching six tetracycline-stained maxillary anterior teeth that have porcelain veneers. *(a)* Prior to bleaching. The patient was unhappy with the gray tones transmitting through the veneers from the underlying tetracycline-stained teeth. *(b)* Palatal view prior to bleaching. The gray tetracycline stain is obvious. *(c)* After 4 months of bleaching with 10% carbamide peroxide in a nonscalloped non-reservoir tray delivered from the lingual, the veneers appear lighter because the underlying teeth are no longer transmitting the gray stain of the tetracycline staining. *(d)* Palatal view after bleaching. The bleaching process has lightened the teeth noticeably.

Restoration of bleached teeth

If micro- or macroabrasion is not successful in removing discolorations after bleaching, or if there are defects or caries present, then a resin composite restoration may be indicated. Placement of the restoration should be delayed at least 1 to 2 weeks for the enamel-composite bond strength to return to normal and for the shade of the tooth to stabilize. These variances are related to the oxygen that enters the tooth from bleaching, and time must be allowed for the oxygen to leave the tooth. After that 1- to 2-week time frame, selecting which teeth need to be restored and the most appropriate treatment is the next step in completing the esthetic outcome.

Because the tooth color may bleach lighter than the traditional lightest shade (B1 on the VITA Classic shade guide) and because all teeth do not bleach to the same endpoint, it is helpful to have multiple shades of composite in small color shifts that are lighter than B1. If there are areas that need to be concealed, such as a white spot that could not be removed, then an opaque material is best, such as an A2 shade (VITA Classic shade guide), with a translucent enamel shade overlying the block-out material of the A2 opaque shade. Stark white tints may be used inside endodontically treated teeth to further lighten them, while pink composite may be helpful to neutralize remaining gray on the tooth from amalgam stains or severe

tetracycline staining or banding.

Because the shade of the resin composite material selected to restore highly bleached teeth will be a lighter shade, additional caution compared with using darker shades must be taken to avoid the operatory light or the room light from prematurely polymerizing the composite prior to the final form being achieved. Premature polymerization can be avoided by using the orange shield typically used to protect eyes from the curing light as a shield from the operatory light. While this bathes the area in orange, it supplies almost unlimited working time with adequate light to see the lingual and occlusal surfaces of the teeth.[265]

The combination of appropriate shade and other esthetic considerations, along with form and function, afford an excellent restoration of a bleached tooth or teeth. See chapter 10 for more information on anterior composite restorations.

Summary

- Whitening is best performed in a professionally supervised manner, with a proper examination and diagnosis, using appropriate materials for the patient and situation, with a fair fee for service.
- Low concentrations of peroxide, especially 10% carbamide peroxide in a custom-fitted tray, are generally the safest, most cost-effective, best-researched whitening treatments available.
- Other bleaching treatments, such as in-office bleaching procedures, may be indicated based on patient preference, lifestyle, finances, or other limitations, but these require informed consent after presenting cost-benefit and risk-benefit ratios.
- OTC nondentist bleaching does not have a good risk-benefit ratio because of the lack of a comprehensive examination to determine teeth staining/darkness etiology. Some products have minimal to no efficacy, while others that do in fact work are not as effective as claimed.

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